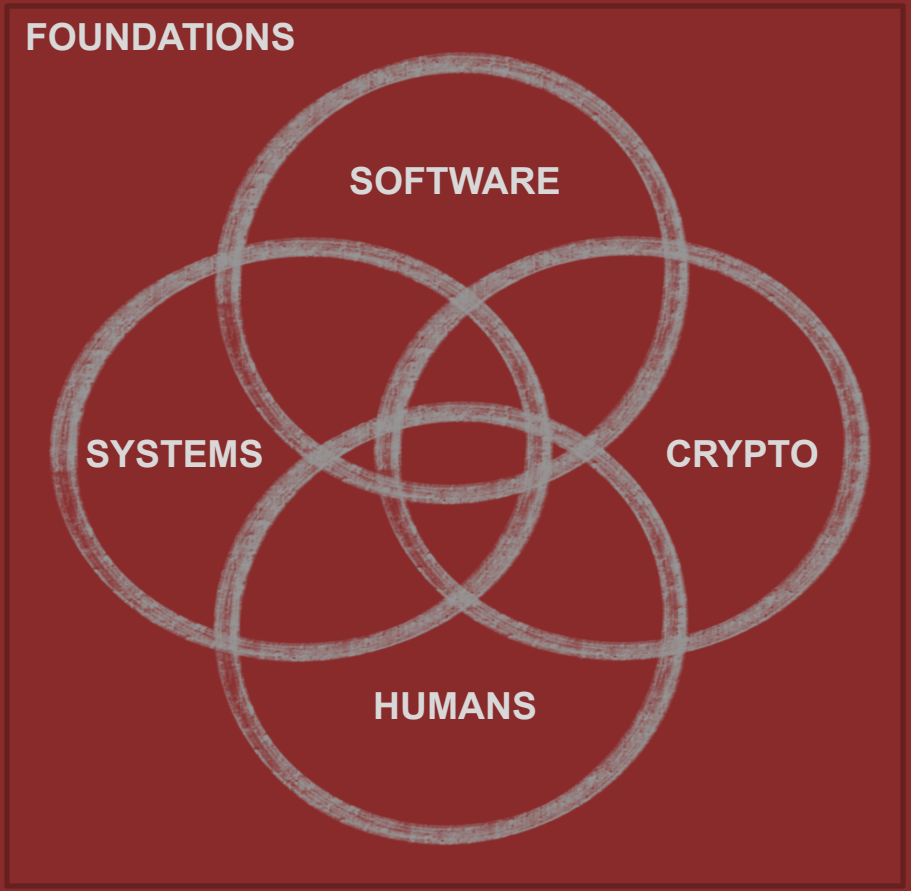
Διάλεξη #2 - x86 Basics and Buffer Overflows

Εθνικό και Καποδιστριακό Πανεπιστήμιο Αθηνών

Εισαγωγή στην Ασφάλεια

Θανάσης Αυγερινός



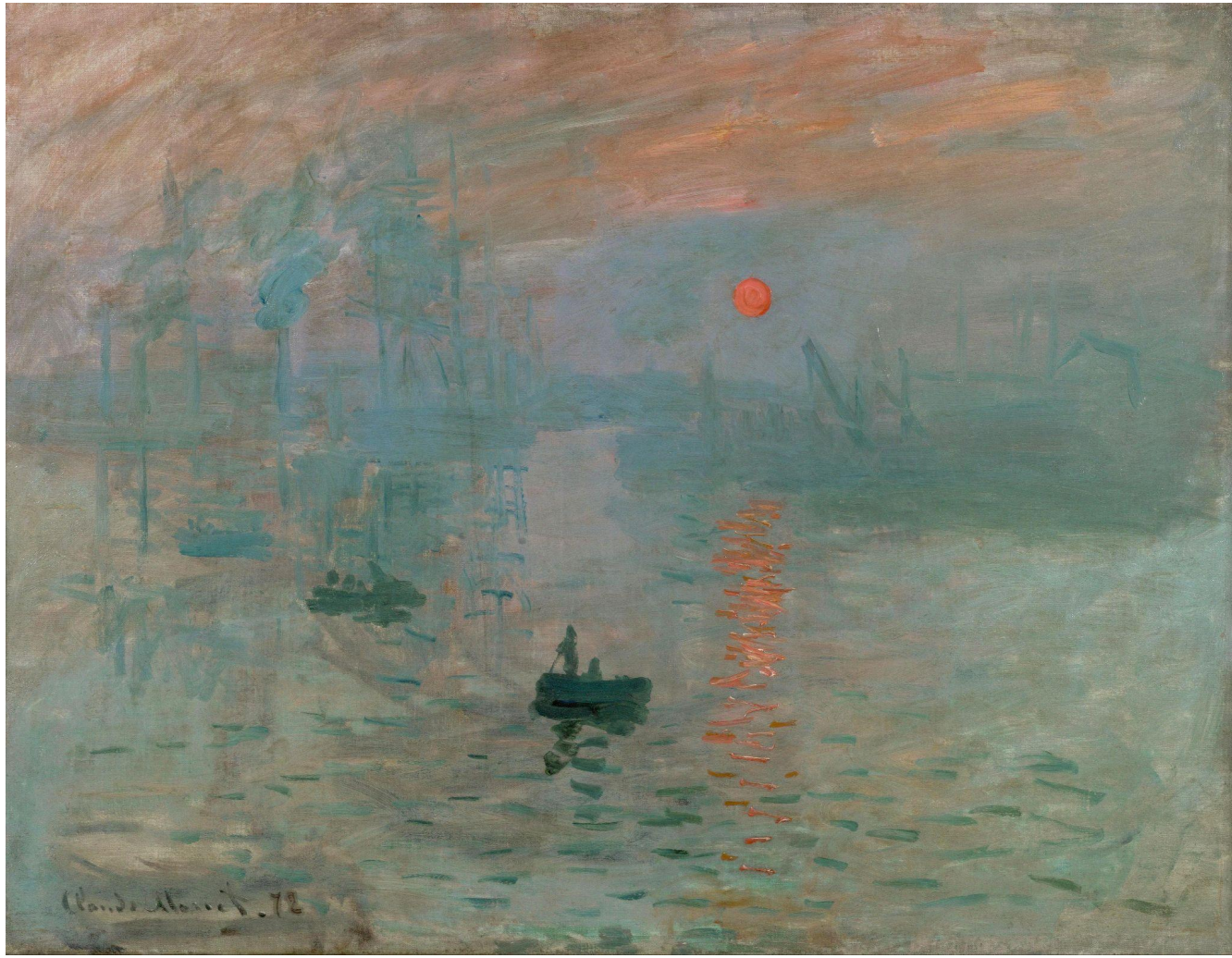
Huge thank you to David Brumley from Carnegie Mellon University for the guidance and content input while developing this class

Ανακοινώσεις / Διευκρινίσεις

- N/A

Την Προηγούμενη Φορά

- Security Fundamentals
  - Adversaries
  - Threat Models
  - Security Properties
  - Trusted Computing Base (TCB)
  - Security Principles



Σήμερα

- x86 Assembly Fundamentals
  - Call Return Semantics
- Basics of buffer overflow attacks

Security in the News

CybersecurityNews

Multiple VMware Aria Vulnerabilities Allow Remote Code Execution Attacks

21 hours ago • By Guru Baran

The Hacker News

CISA Confirms Active Exploitation of FileZen CVE-2026-25108 Vulnerability

1 hour ago • By Ravie Lakshmanan

Google Cloud

Diverse Threat Actors Exploiting Critical WinRAR Vulnerability CVE-2025-8088

Jan 27

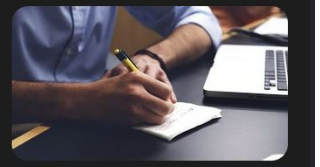
Unit 42

VShell and SparkRAT Observed in Exploitation of BeyondTrust Critical Vulnerability (CVE-2026-1731)

6 days ago

Zero Day Initiative

CVE-2026-20841: Arbitrary Code Execution in the Windows Notepad



6 days ago

Phoronix

CVE-2026-0915: GNU C Library Fixes A Security Issue Present Since 1996

Jan 17 • By Michael Larabel

Which one to look at first?!

Remote Code Execution Flaw In Grandstream GXP1600 Phones Exposed With PoC Exploit

BY VARSHINI / FEBRUARY 24, 2026 /

A critical zero-day vulnerability in Grandstream's GXP1600 series VoIP phones, designated CVE-2026-2329, allowing unauthenticated remote code execution with root privileges.

This stack-based buffer overflow affects the web-based API service on TCP port 80, accessible in default configurations without any authentication.

All six models GXP1610, GXP1615, GXP1620, GXP1625, GXP1628, and GXP1630 share the same firmware image and are vulnerable in versions up to 1.0.7.79.

The flaw carries a CVSSv4 score of 9.3 (Critical), stemming from CWE-121: Stack-based Buffer Overflow, enabling network-accessible attacks with low complexity.

Grandstream released firmware version 1.0.7.81 on January 30, 2026, explicitly noting fixes for security vulnerabilities that fully remediate the issue.

Organizations using these entry-level IP phones in small businesses or enterprises face high risks, as the devices often handle sensitive voice communications over SIP protocols.

In the gs_web binary (32-bit ARM Little Endian), code iterates over this parameter, appending characters to a fixed 64-byte stack buffer named small_buffer without bounds checking.

When the buffer overflows triggered by inputs exceeding 63 characters plus null it corrupts adjacent stack memory, including the program counter (PC), as confirmed via GDB debugging and core dumps.

CVE ID	CVSS Score	Affected Versions	Patched Version	Models
CVE-2026-2329	9.3 (Critical)	≤1.0.7.79	1.0.7.81	GXP1610, GXP1615, GXP1620, GXP1625, GXP1628, GXP1630

Security mitigations are weak, no RELRO, canaries, PIE, or fortify_source, though the NX bit prevents direct stack execution.

Exploitation uses return-oriented programming (ROP) chains, leveraging the non-PIE load address at 0x00008000 for predictable gadgets. Multiple overflows via colons allow precise placement of null bytes, overcoming null restrictions for ROP payloads that call system() and exit().

Any real impact though? What do *you* think?

Remote Code Execution Flaw In Grandstream GXP1600 Phones Exposed With PoC Exploit

BY VARSHINI / FEBRUARY 24, 2026 /

What is all that?! 👇

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```
Core was generated by `/app/bin/gs_web'.
Program terminated with signal SIGSEGV, Segmentation fault.
#0  0x41414140 in ?? ()
gef➢  i r
r0             0xceefe814          0xceefe814
r1             0x8                 0x8
r2             0x23180             0x23180
r3             0x22514             0x22514
r4             0x41414141          0x41414141
r5             0x41414141          0x41414141
r6             0x41414141          0x41414141
r7             0x41414141          0x41414141
r8             0x41414141          0x41414141
r9             0x41414141          0x41414141
r10            0x41414141          0x41414141
r11            0x41414141          0x41414141
r12            0x222a0             0x222a0
sp             0xceefe898          0xceefe898
lr             0x14468             0x14468
pc             0x41414140          0x41414140
cpsr           0x60000030          0x60000030
gef➢  bt
#0  0x41414140 in ?? ()
#1  0x00014468 in ?? ()
#2  0x41414140 in ?? ()
Backtrace stopped: previous frame identical to this frame (corrupt stack?)
gef➢  x/32x $pc
0x41414140:     Cannot access memory at address 0x41414140
gef➢  x/32x $sp
0xceefe898:     0x41414141      0x41414141      0x41414141      0x41414141
0xceefe8a8:     0x41414141      0x41414141      0x41414141      0x41414141
0xceefe8b8:     0x41414141      0x41414141      0x41414141      0x41414141
0xceefe8c8:     0x41414141      0x41414141      0x41414141      0x41414141
0xceefe8d8:     0x41414141      0x41414141      0x41414141      0x41414141
0xceefe8e8:     0x41414141      0x41414141      0x41414141      0x41414141
0xceefe8f8:     0x41414141      0x41414141      0x41414141      0x41414141
0xceefe908:     0x41414141      0x41414141      0x41414141      0x41414141
```

Reminders: Memory, Stack, and Endianness (speedrunning progintro)

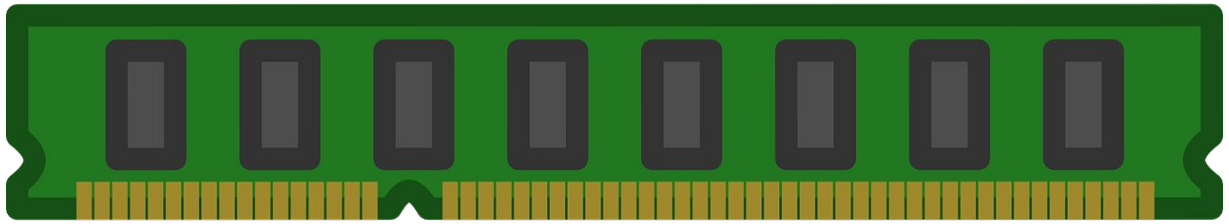
Η Μνήμη Οργανώνεται σε Bytes (Υπενθύμιση)

Το μέγεθος της μνήμης μετράται σε Bytes:

- **1 KB (KiloByte) = 1.000 Bytes**
- **1 MB (MegaByte) = 1.000.000 Bytes**
- **1 GB (GigaByte) = 1.000.000.000 Bytes**

Μνήμη με N Bytes

Byte 0	0	0	1	0	1	0	1	0
Byte 1	0	1	0	0	0	0	1	0
Byte 2	1	1	1	0	0	0	1	1
…				…				
Byte N-1	0	0	0	0	0	0	0	0
Byte N	0	1	1	0	1	0	0	0



Κατηγορίες Μνήμης

Υπάρχουν 3 κατηγορίες μνήμης:

1. Η στοίβα (stack)
2. Ο σωρός (heap)
3. Η παγκόσμια / στατική μνήμη (global / static memory)

Η στοίβα (Stack)

Η στοίβα (stack) είναι μια συνεχόμενη περιοχή της μνήμης που προστίθενται και αφαιρούνται στοιχεία με σειρά Last-In-First-Out (LIFO), δηλαδή το τελευταίο στοιχείο που προστέθηκε είναι το πρώτο που θα αφαιρεθεί.

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```
char a = 61; char b = 62; char c = 63;
```

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char a = 61; char b = 62; char c = 63;
...
 char d = 64;
```

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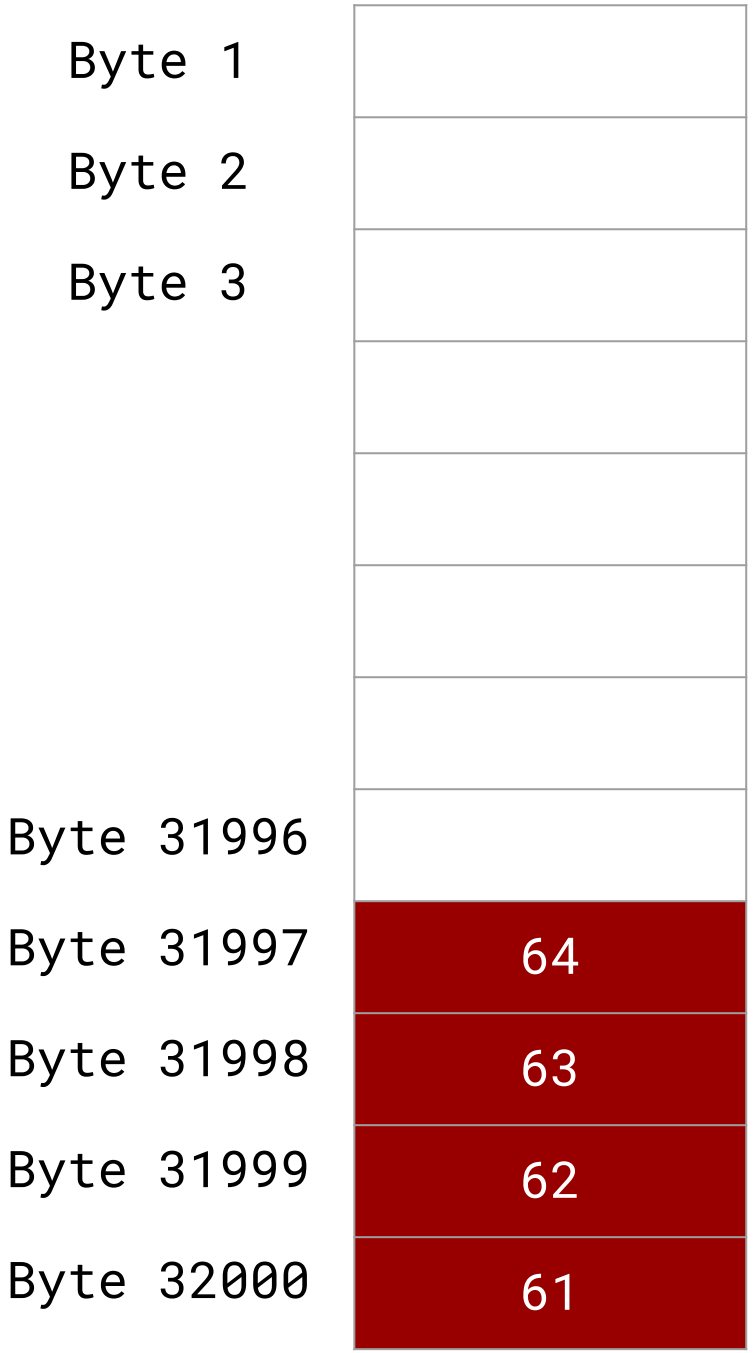

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Τι αποθηκεύεται συνήθως στην στοίβα;

2: Ορίσματα που περνάμε στην συνάρτηση

```
int equalIgnoreCase(char char1, char char2) {
    char lowerChar1 = tolower(char1);
    char lowerChar2 = tolower(char2);
    return lowerChar1 == lowerChar2;
}
```

1: Τοπικές Μεταβλητές ορισμένες μέσα στην συνάρτηση

3: Προσωρινά δεδομένα (συνήθως μερικά bytes) που αποθηκεύει ο μεταγλωττιστής

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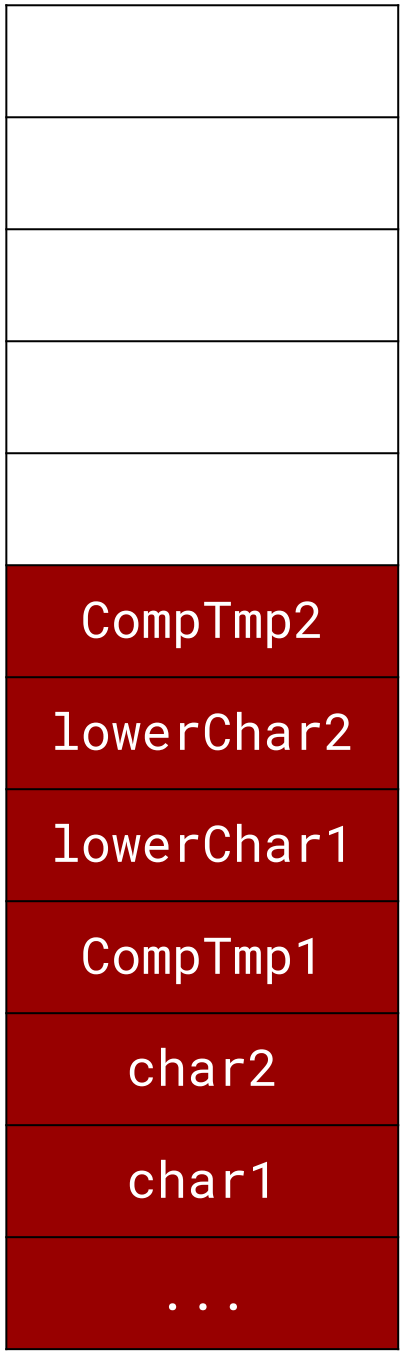

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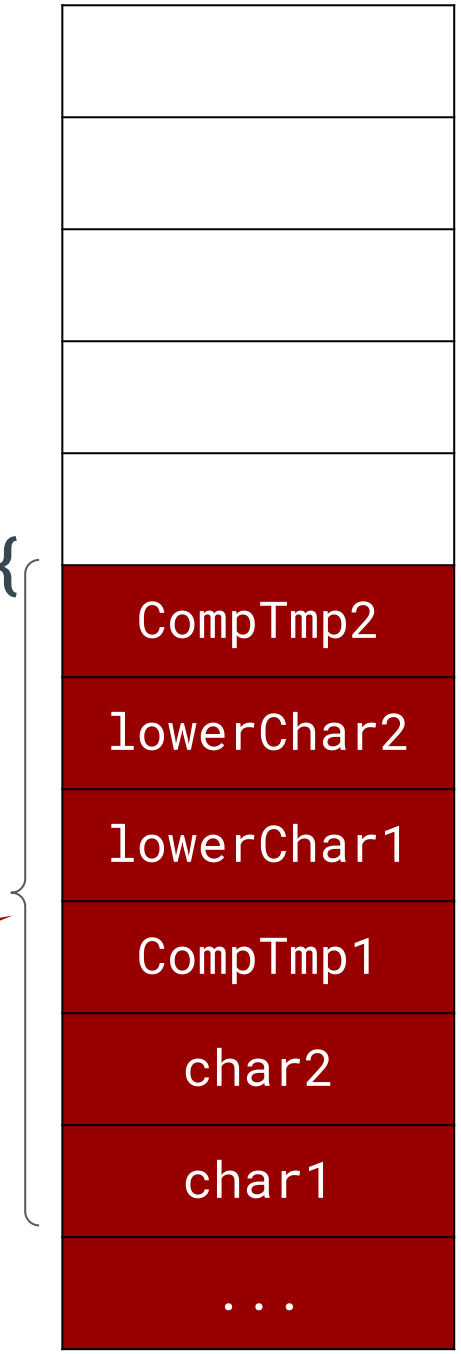

Χώρος που δεσμεύεται στην στοίβα σε **κάθε** κλήση της συνάρτησης equalIgnoreCase. Αυτό το κομμάτι μνήμης λέγεται και διάγραμμα ενεργοποίησης (**activation record ή stack frame**) της συνάρτησης

Τι αποθηκεύεται συνήθως στην στοίβα;

```
char tolower(char c) {
    if (c >= 'A' && c <= 'Z')
        return c + ('a' - 'A');
    return c;
}
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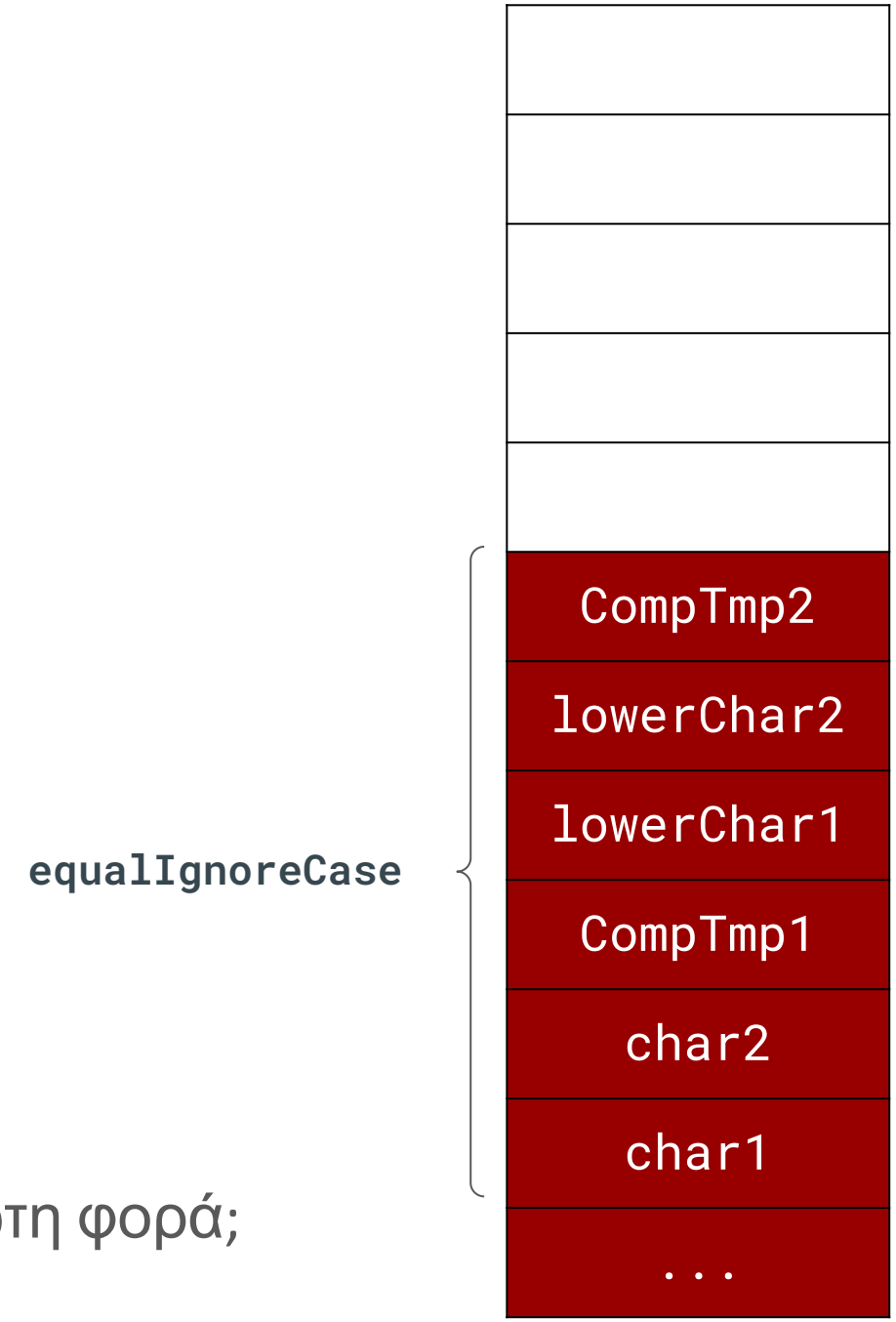

Τι θα συμβεί όταν κληθεί η συνάρτηση tolower την πρώτη φορά;

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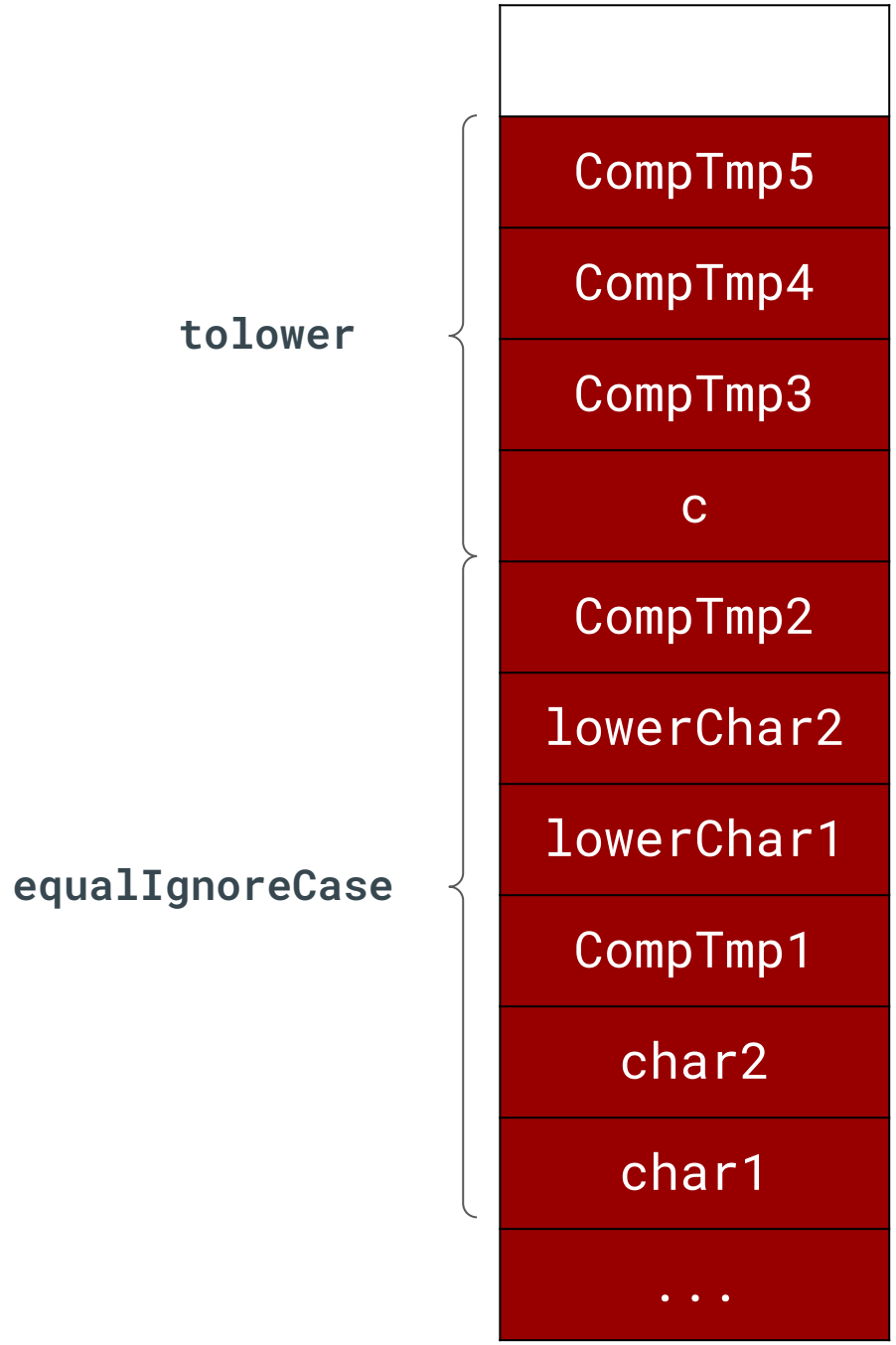

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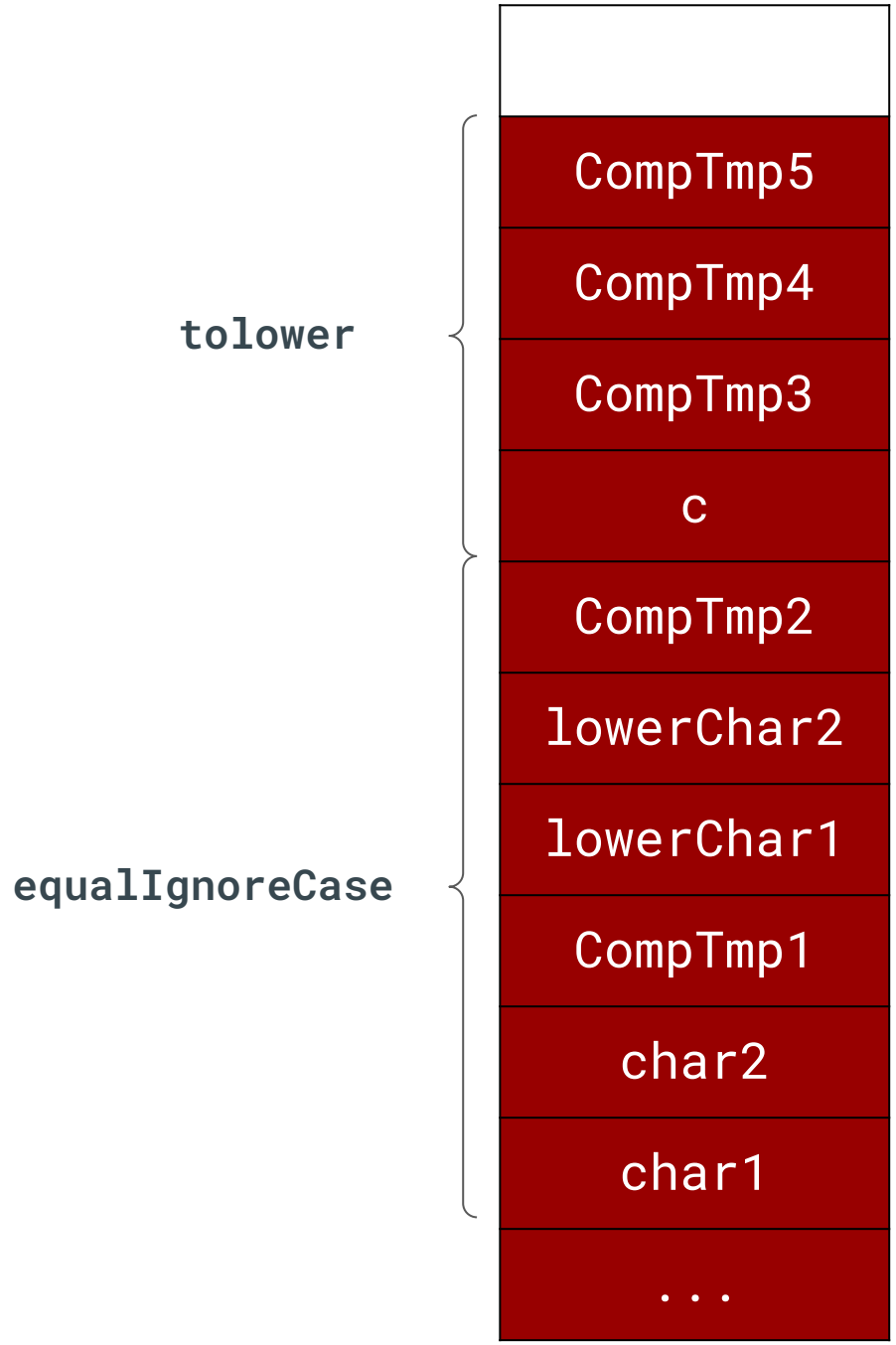

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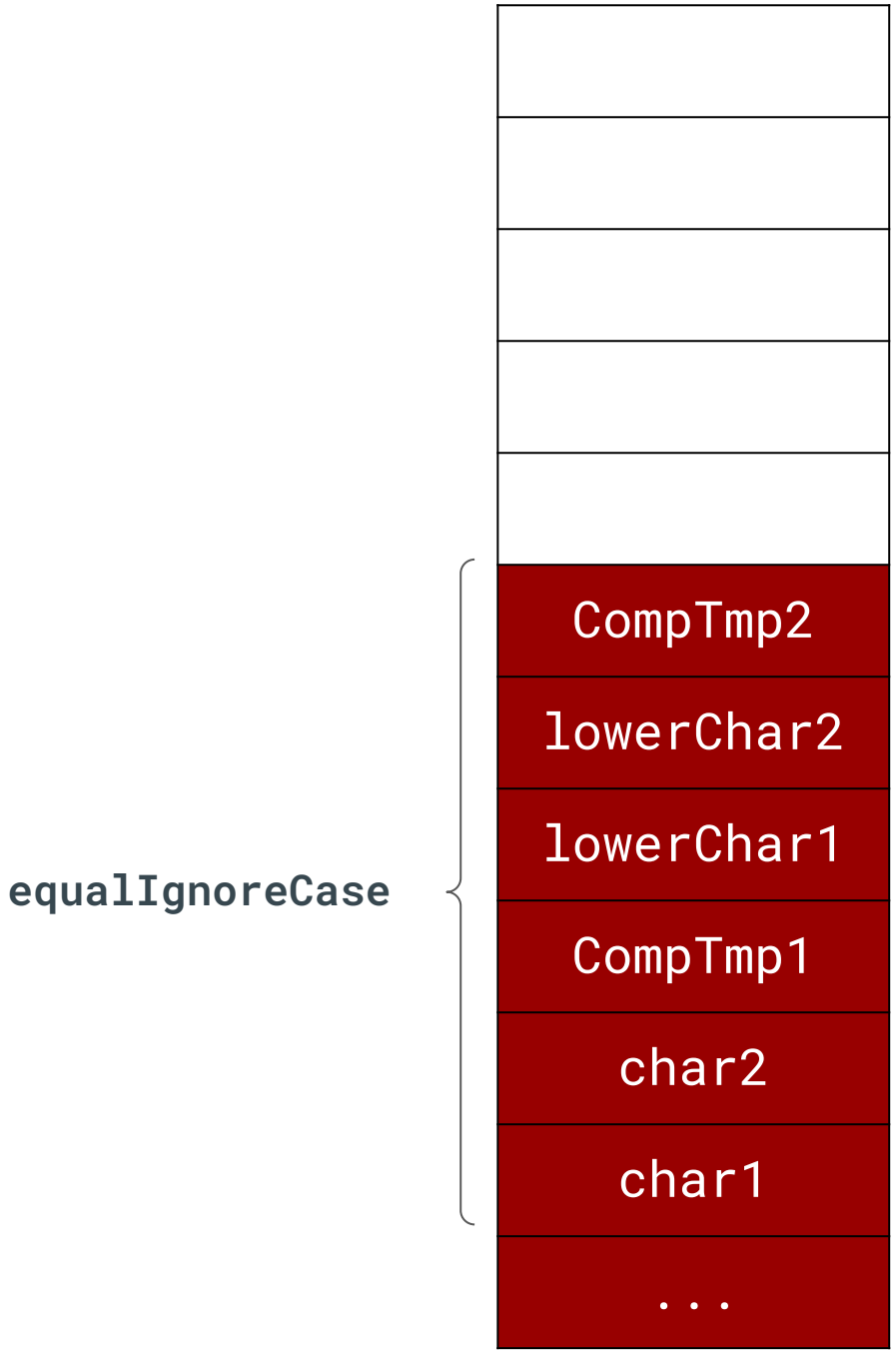

Τι θα συμβεί όταν εκτελεστεί η δεύτερη κλήση tolower;

Τι αποθηκεύεται συνήθως στην στοίβα;

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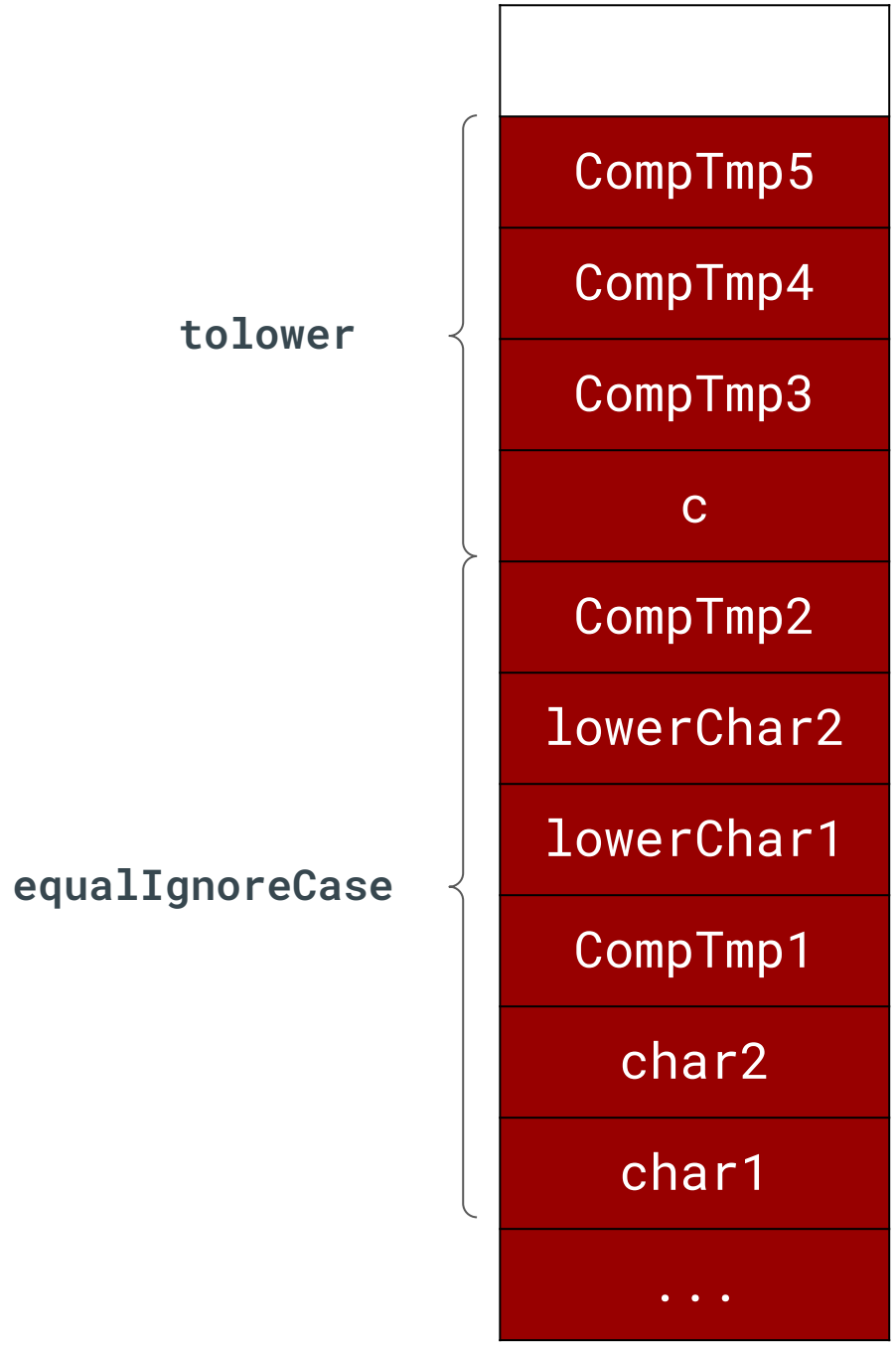

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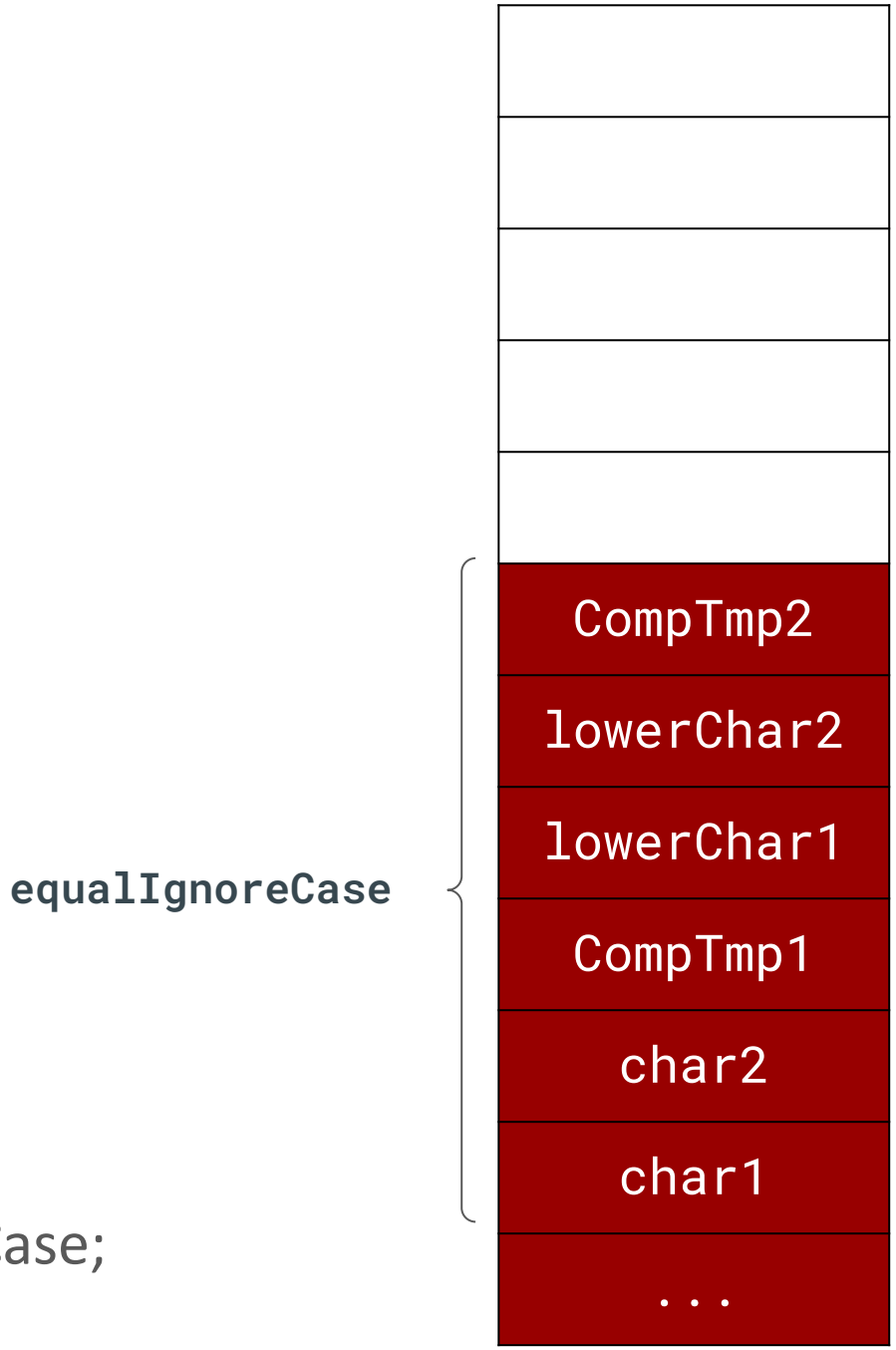

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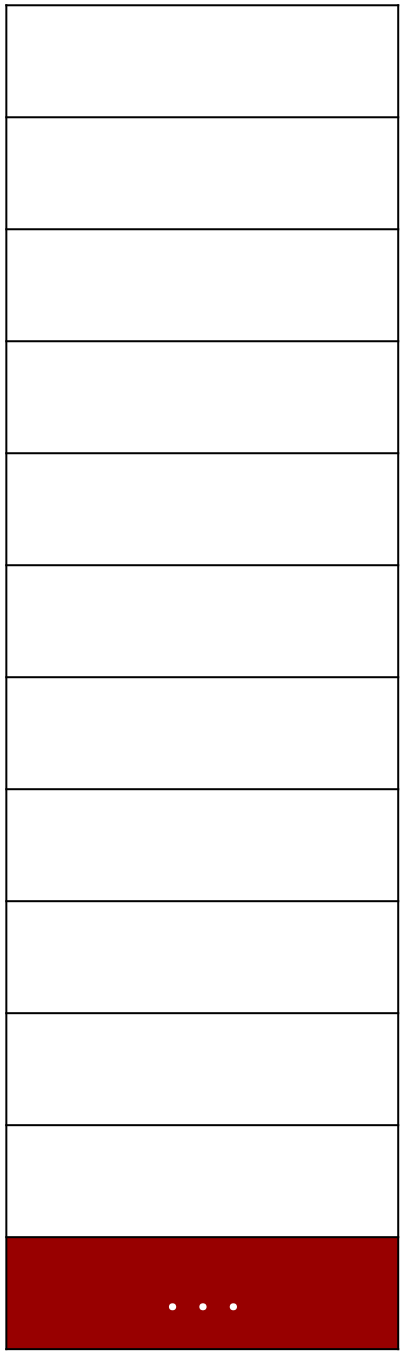

Πόσα bytes έχει ένας int;

Endianness

Endianness λέμε τον τρόπο με τον οποίο οι ακέραιοι αποθηκεύονται στην μνήμη. Οι ακέραιοι αποτελούνται από πολλά bytes και επομένως πρέπει να αποφασίσουμε αν τους αποθηκεύουμε από το μικρότερο στο μεγαλύτερο (little endian) ή από το μεγαλύτερο στο μικρότερο (big endian).

68 ('h')	65 ('e')	6c ('l')	6c ('l')

Το πρώτο byte στην μνήμη είναι το μικρότερο byte του αριθμού

6c6c6568

Παράδειγμα Endianness

Τι θα τυπώσει το παρακάτω:

```
#include <stdio.h>
int main() {
  int x = 0x42434445;
  char * bytes = (char*)&x;
  int i;
  for(i = 0; i < sizeof(int) / sizeof(char); i++)
   printf("%02x\n", bytes[i]);
  return 0;
}
```

Παράδειγμα Endianness

Τι θα τυπώσει το παρακάτω:

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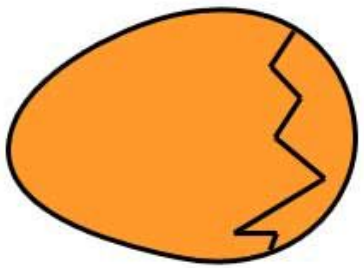
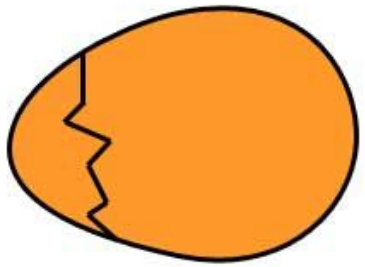
int main() {
  int x = 0x42434445;
  char * bytes = (char*)&x;
  int i;
  for(i = 0; i < sizeof(int) / sizeof(char); i++)
   printf("%02x\n", bytes[i]);
  return 0;
}
```

```
$ ./int
45
44
43
42
```

ENDIANNESS



BIG ENDIAN - The way people always broke their eggs in the Lilliput land



LITTLE ENDIAN - The way the king then ordered the people to break their eggs

- Little endian (x86/x86_64/arm/...): LSB stored in smallest memory address
- Big endian (PPC/SPARC/...): MSB stored in smallest memory address
- Bi-endian (Some arm and mips): Can switch dynamically

Credit: http://flickeringtubelight.net/blog/2004/05/big-endian-and-little-endian-storage-schemes-how-to-remember/

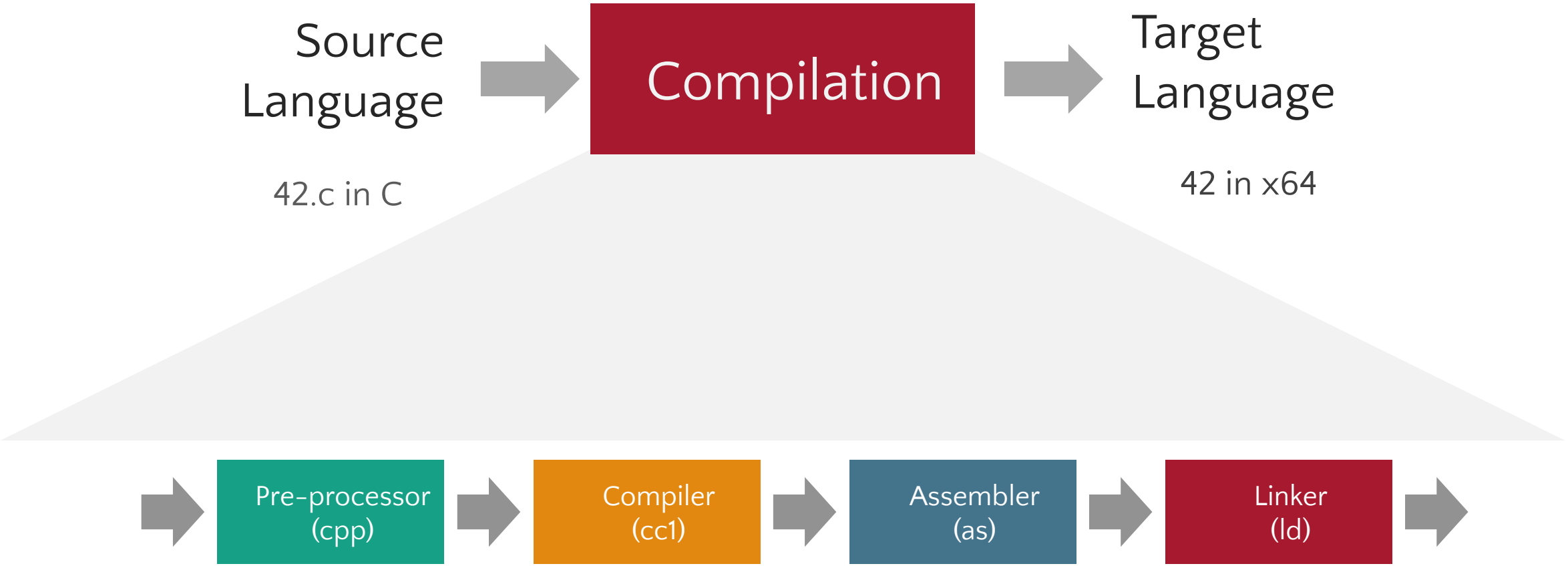
Compilation Workflow & Execution Semantics

To answer

Is this program safe?

We must first know

What will executing it do?


Source Language
42.c in C
Compilation
Target Language
42 in x64
Pre-processor (cpp)
Compiler (cc1)
Assembler (as)
Linker (ld)

Preprocessor

Compiler

Assembler

Linker

```
$ gcc -E
```

```c
#include <stdio.h>

void answer_function(char *name, int x)
{
  printf("% s answered % d\n", name, x);
}

int main(int argc, char *argv[])
{
  answer_function("hello", 42);
}
```

#include expansion

#define substitution

Preprocessor

Compiler

Assembler

Linker

```
$ gcc -S
```

```
answer_function:
.LFB0:
.cfi_startproc
pushq %rbp
.cfi_def_cfa_offset 16
.cfi_offset 6, -16
movq %rsp, %rbp
.cfi_def_cfa_register 6
subq $16, %rsp
movq %rdi, -8(%rbp)
movl %esi, -12(%rbp)
movl -12(%rbp), %edx
```

Generate assembly code

Preprocessor

Compiler

Assembler

Linker

```
$ as hello.s -o hello.o
$ file hello.o
hello.o: ELF 64-bit LSB relocatable,
x86-64, version 1 (SYSV), not stripped
```

Create object code

(the ones and zeros a computer executes)

Preprocessor

Compiler

Assembler

Linker

$ ld <opts>
(this one is complicated; see gcc -v)

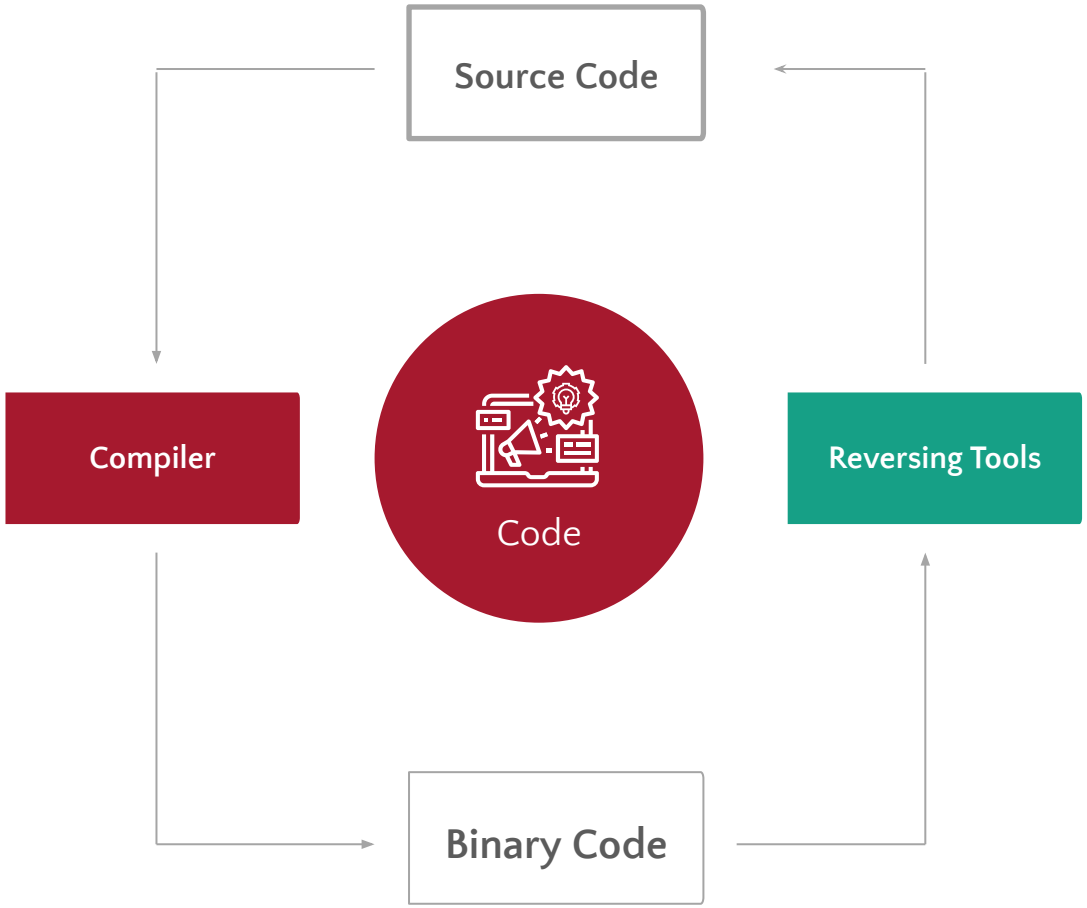
```
gcc -v hello.c
 ...
/usr/lib/gcc/x86_64-linux-gnu/10/collect2 -plugin
/usr/lib/gcc/x86_64-linux-gnu/10/liblto_plugin.so
-plugin-opt=/usr/lib/gcc/x86_64-linux-gnu/10/lto-wrapper
-plugin-opt=-fresolution=/tmp/ccLvq7Mj.res -plugin-opt=-pass-through=-lgcc
-plugin-opt=-pass-through=-lgcc_s -plugin-opt=-pass-through=-lc
-plugin-opt=-pass-through=-lgcc -plugin-opt=-pass-through=-lgcc_s --build-id --eh-frame-hdr
-m elf_x86_64 --hash-style=gnu --as-needed -dynamic-linker /lib64/ld-linux-x86-64.so.2 -pie
/usr/lib/gcc/x86_64-linux-gnu/10/../../../x86_64-linux-gnu/Scrt1.o
/usr/lib/gcc/x86_64-linux-gnu/10/../../../x86_64-linux-gnu/crti.o
/usr/lib/gcc/x86_64-linux-gnu/10/crtbeginS.o -L/usr/lib/gcc/x86_64-linux-gnu/10
-L/usr/lib/gcc/x86_64-linux-gnu/10/../../../x86_64-linux-gnu
-L/usr/lib/gcc/x86_64-linux-gnu/10/../../../../lib -L/lib/x86_64-linux-gnu -L/lib/../lib
-L/usr/lib/x86_64-linux-gnu -L/usr/lib/../lib -L/usr/lib/gcc/x86_64-linux-gnu/10/../../..
/tmp/ccRnv5wm.o -lgcc --push-state --as-needed -lgcc_s --pop-state -lc -lgcc --push-state
--as-needed -lgcc_s --pop-state /usr/lib/gcc/x86_64-linux-gnu/10/crtendS.o
/usr/lib/gcc/x86_64-linux-gnu/10/../../../x86_64-linux-gnu/crtn.o
```

Linker links with other compilation units (.o) and libraries (.a) to produce an executable.

Compilation is not necessarily invertible

Compilation

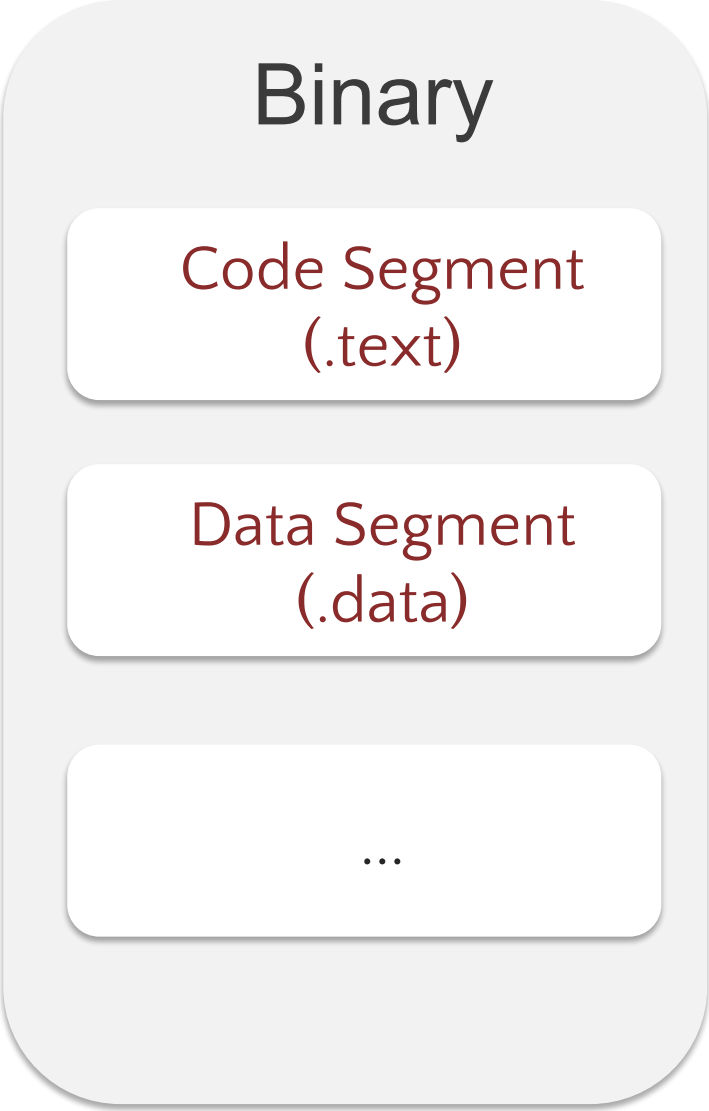
Turn source code into executable code using programmer-defined abstractions as a guide translation.



Reversing

Best case effort to recover source code from binary code.

Interview Question: How would you see the content of a binary program?



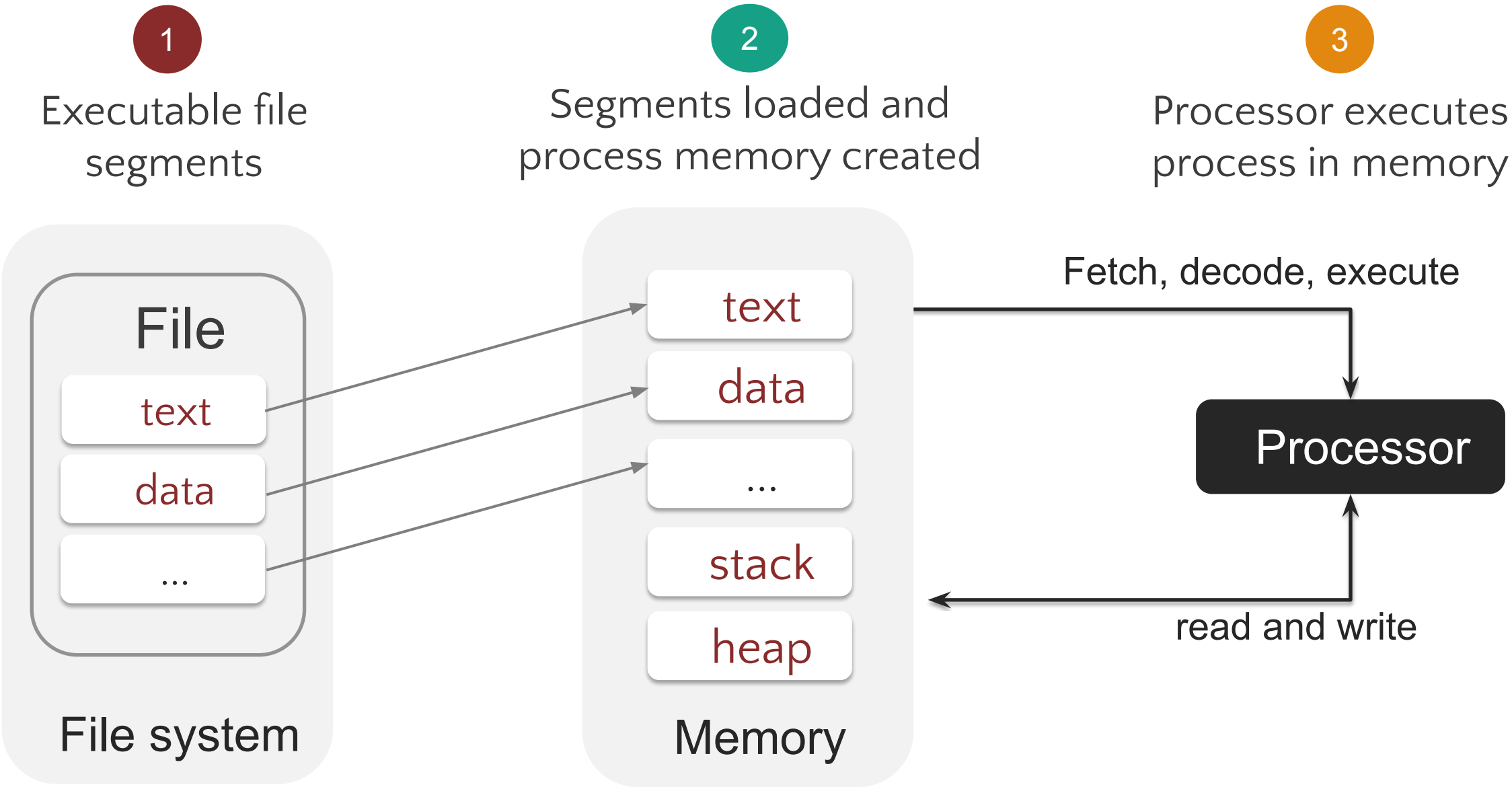
The final **executable binary**

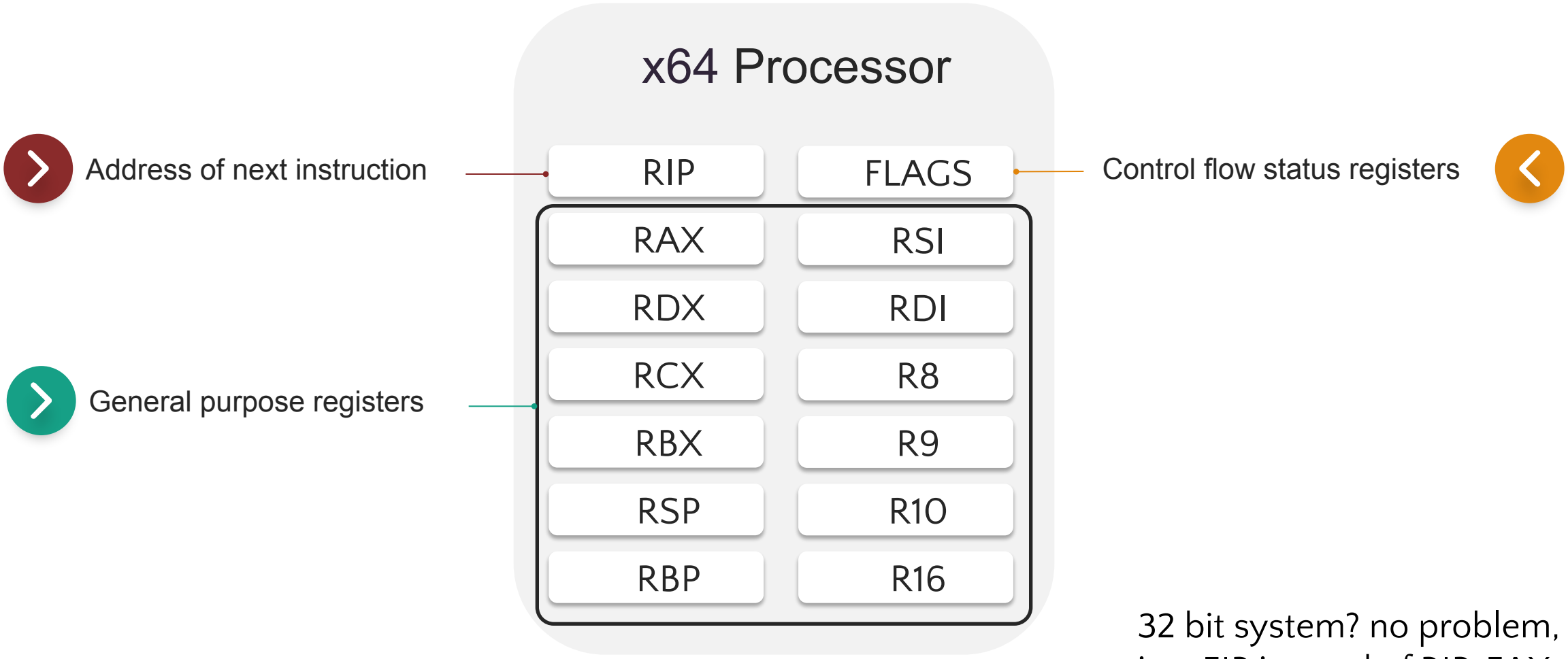
- Text segment
- Data for constants like “hello world” and globals
- And more...

```
# read the various sections. Also see `nm`
$ readelf -S <file>

# -d: disassemble
# -S: match symbol locations to source
$ objdump <-d> <-S> file
```

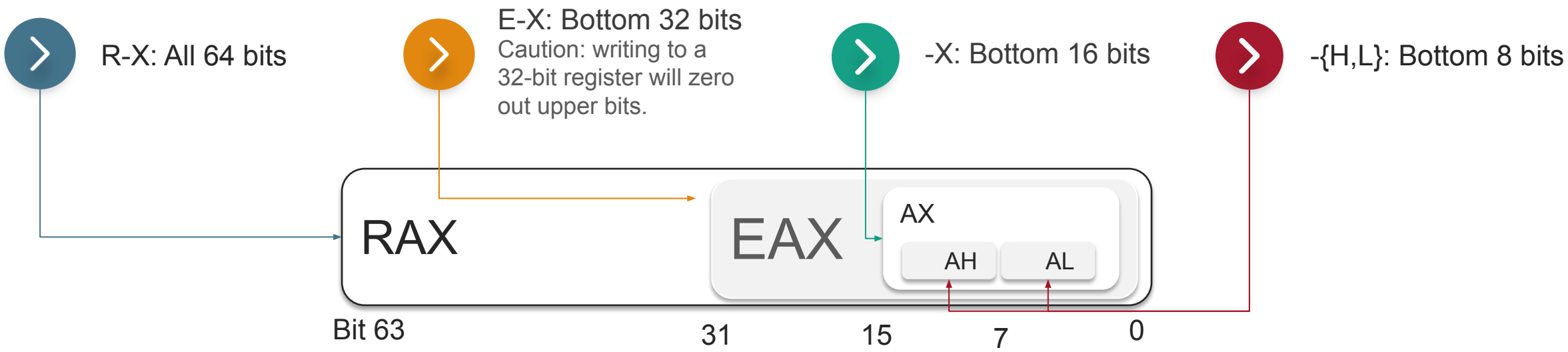
BASIC EXECUTION MODEL


1
Executable file segments
2
Segments loaded and process memory created
3
Processor executes process in memory
File
text
data
...
File system
text
data
...
stack
heap
Memory
Fetch, decode, execute
Processor
read and write



32 bit system? no problem, just EIP instead of RIP, EAX instead of RAX, etc

Parts of a register can be addressed directly



Basic Ops and AT&T Vs. Intel Syntax

Meaning	AT&T	Intel
rbx = rax	movq %rax, %rbx	mov rbx, rax
rax = rax + rbx	addq %rbx, %rax	add rax, rbx
rcx = rcx << 2	shl $2, %rcx	shl rcx, 2

AT&T is at odds w/ assignment order

- Default for objdump
- Default in UNIX

Intel mirrors assignment order

- objdump –M intel
- Default for windows

Memory Operations

Memory Loads

What do these instructions do?

```
1.  movb (%rax), %dl

2.  mov (%rax), %edx
```

src dst

- Bracket "[]" indicates Intel
- Paren "()" indicates AT&T

Register	Value
rax	0x3
rdx	0x0
rbx	0x5

Byte	Address
0xff	6
0xee	5
0xdd	4
0xcc	3
0xbb	2
0xaa	1
0x00	0

Memory Loads

What do these instructions do?

```
1.  movb (%rax), %dl

1.  mov (%rax), %edx
```

Register	Value
rax	0x3
rdx	0x0
rbx	0x5

Byte	Address
0xff	6
0xee	5
0xdd	4
0xcc	3
0xbb	2
0xaa	1
0x00	0

1. Moves 0xcc into dl
2. Moves 0xcc – 0xff into edx.
   But what number is this?

EXAMPLE

We'll assume x86/x86_64 from now on:

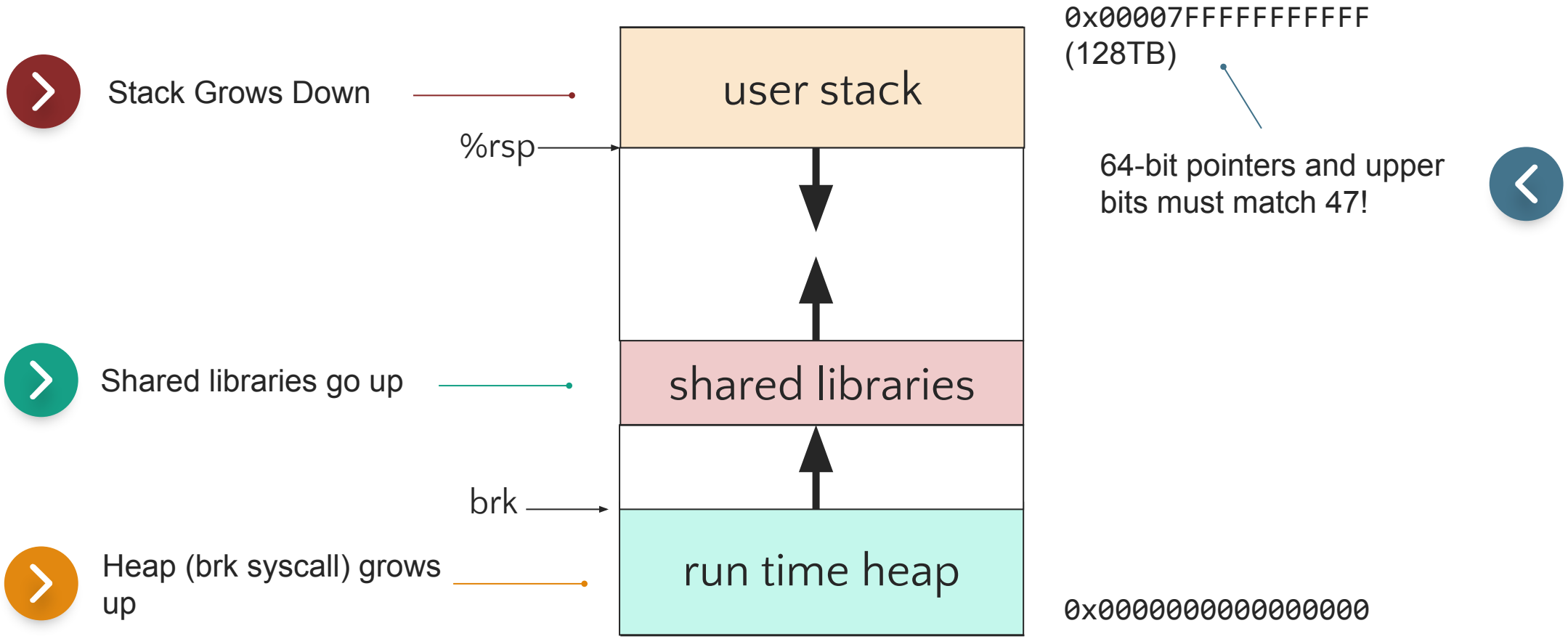
- Address *a* goes into register bits 0-7
- Address *a+1* in the 8-15
- And so on

```
mov (%rax), %edx
```

Register	Value
rax	0x3
rdx	0xcc
rbx	0x5

	Address
0xff	6
0xee	5
0xdd	4
0xcc	3
0xbb	2
0xaa	1
0x00	0

Process Memory Organization

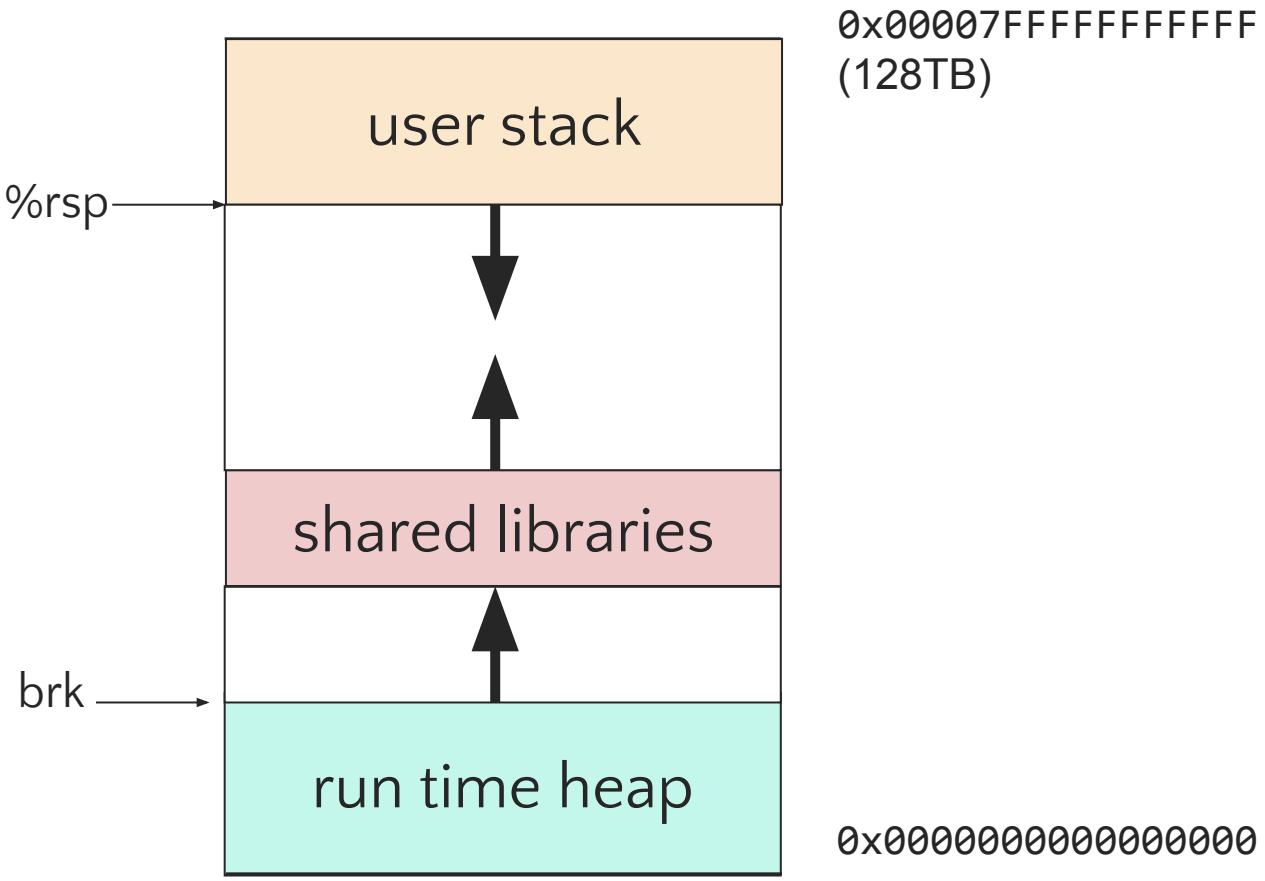

Stack Grows Down
%rsp
user stack
0x00007FFFFFFFFFFF
(128TB)
64-bit pointers and upper bits must match 47!
Shared libraries go up
shared libraries
brk
Heap (brk syscall) grows up
run time heap
0x0000000000000000

On the stack

- Local variables
- Lifetime:
  stack frame

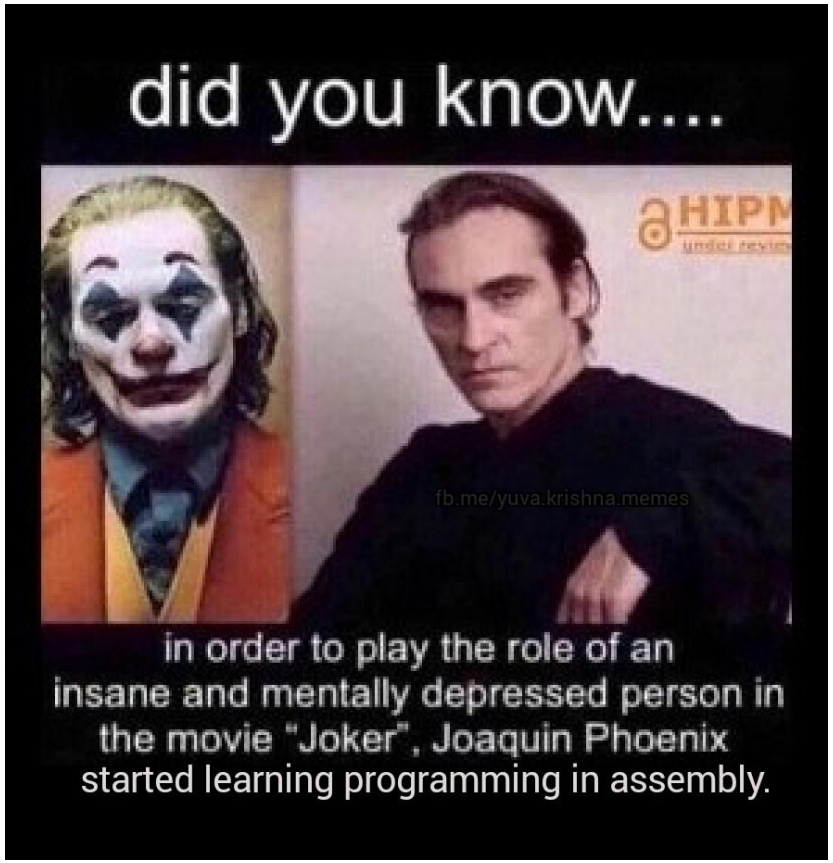
On the heap

- Dynamically allocated
  via new/malloc/etc.
- Lifetime:
  until freed



Procedures

- Assembly code doesn't have procedures
- Compilers *implement* procedures
  - On the stack
  - Following the call/return stack discipline



The compiler needs to implement a call stack, calling convention, and a way to pass arguments for code being compiled, third-party binary libraries, and the OS

```
int f1(int a, int b)
{
  char buf[16];
  int c, d;
  if(a > b)
     c = a;
  else
     c = b;
  d = f2(c, buf);
  return d;
}
```

- Need to access arguments
- Need space to store local variables (buf, c, and d)
- Need space for call arguments
- Need a place to return values

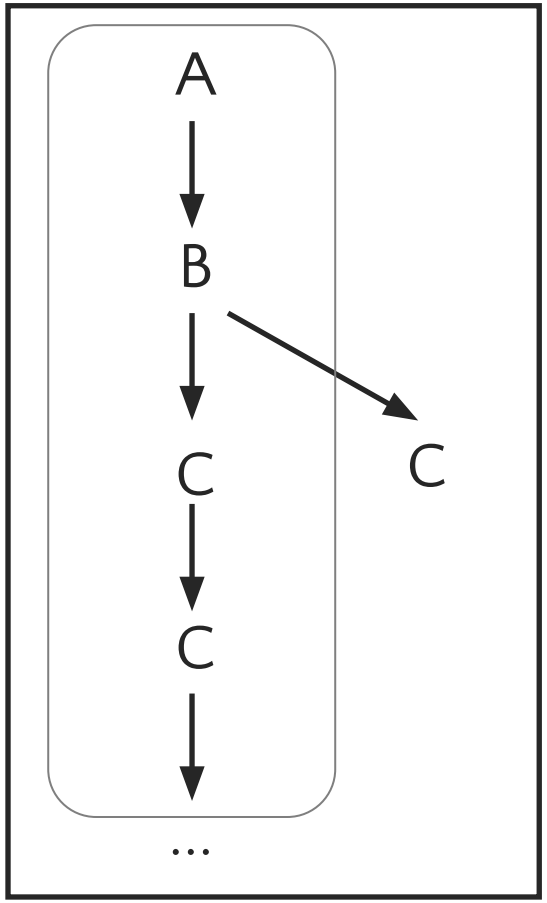
Recall the basic stack semantics of function calls

```
A(...)
{
   ...
   B()
   ...
}
```

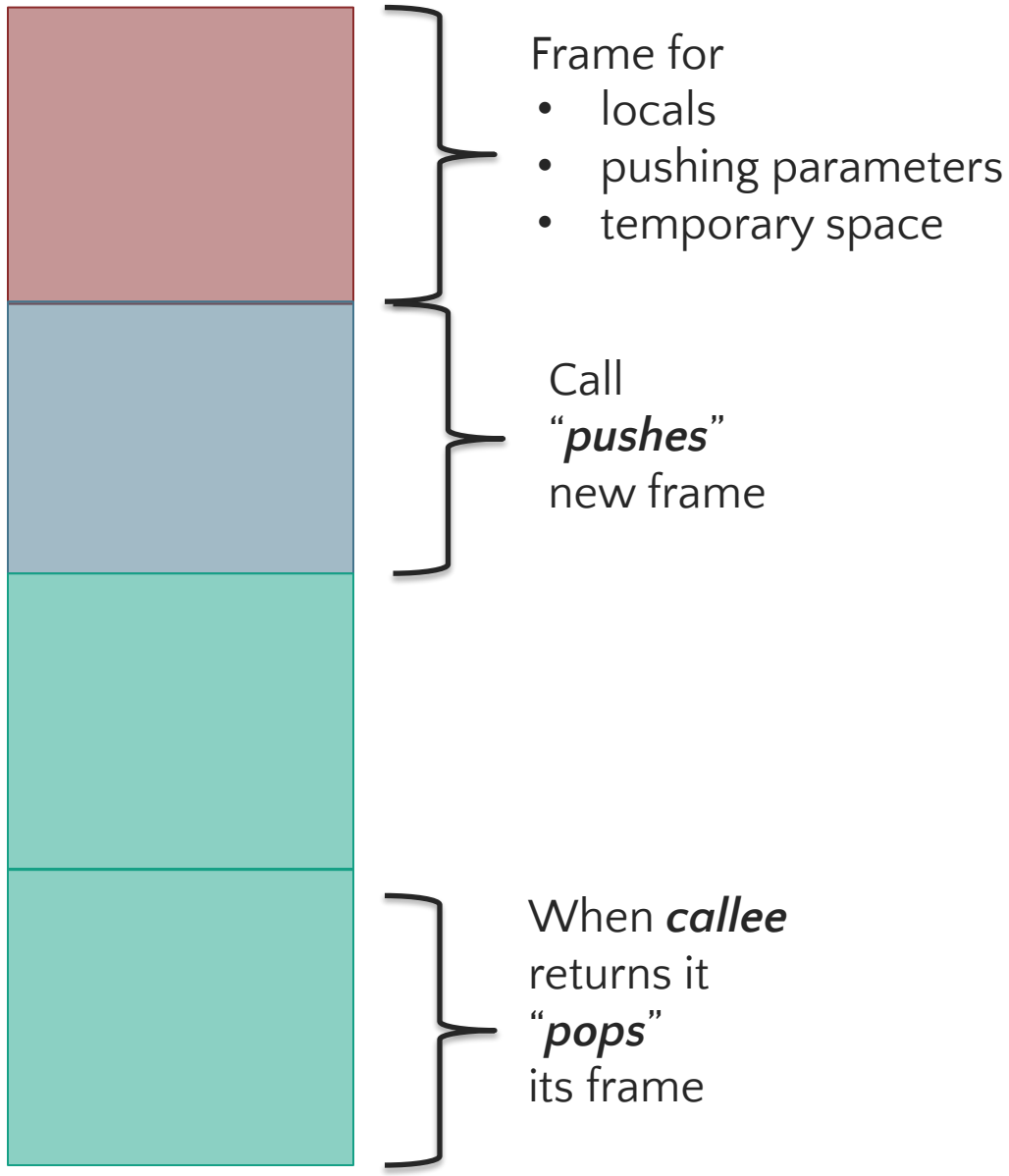
```
B(...)
{
   ...
   C()
   ...
   C()
}
```

```
C(...)
{
   ...
   C()
   ...
}
```

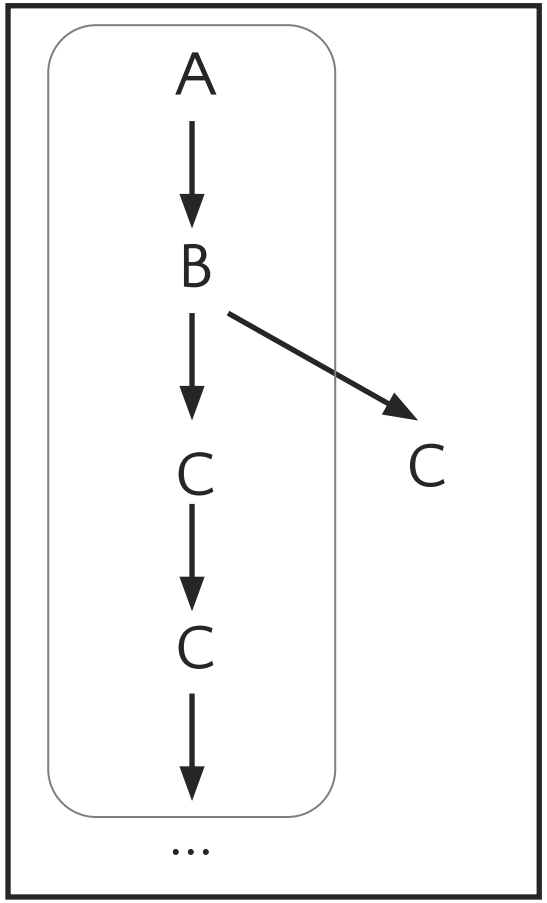
Function Call Chain



The Stack



Function Call Chain



A Toy Example

```
void print(char * message) {
    char buffer[64];
    strcpy(buffer, message);
    printf("%s\n", buffer);
}
int main(int argc, char ** argv) {
    ...
    print(argv[1]);
    return 0;
}
```

A Toy Example

```c
void print(char * message) {
    char buffer[64];
    strcpy(buffer, message);
    printf("%s\n", buffer);
}
int main(int argc, char ** argv) {
    ...
    print(argv[1]);
    return 0;
}
```

```
080491d7 <print>:
...
 80491f2:       e8 69 fe ff ff    call 8049060 <strcpy@plt>
 80491f7:       83 c4 10          add $0x10,%esp
...
 8049202:       e8 69 fe ff ff    call 8049070 <puts@plt>
 8049207:       83 c4 10          add $0x10,%esp
...
 804920f:       c3                ret

08049210 <main>:
...
 804925e:       e8 74 ff ff ff    call 80491d7 <print>
 8049263:       83 c4 10          add $0x10,%esp
...
 8049274:       c3                ret
```

Return to the caller using the `ret` instruction

Transfer control to functions using the `call` instruction

Call - Ret Semantics

Call semantics: Store address of next instruction (the return address) where the current stack pointer is pointing and then jump to target address. call target is equivalent to:

push next_address

jmp target

mem[sp] = next_addr; sp -= sizeof(next_addr)

Ret semantics: Read address of next instruction from the stack (the return address) and jump to it. ret is equivalent to:

pop next_address

jmp next_address

next_addr = mem[sp]; sp += sizeof(next_addr)

A Toy Example

```c
void print(char * message) {
    char buffer[64];
    strcpy(buffer, message);
    printf("%s\n", buffer);
}
int main(int argc, char ** argv) {
    ...
    print(argv[1]);
    return 0;
}
```

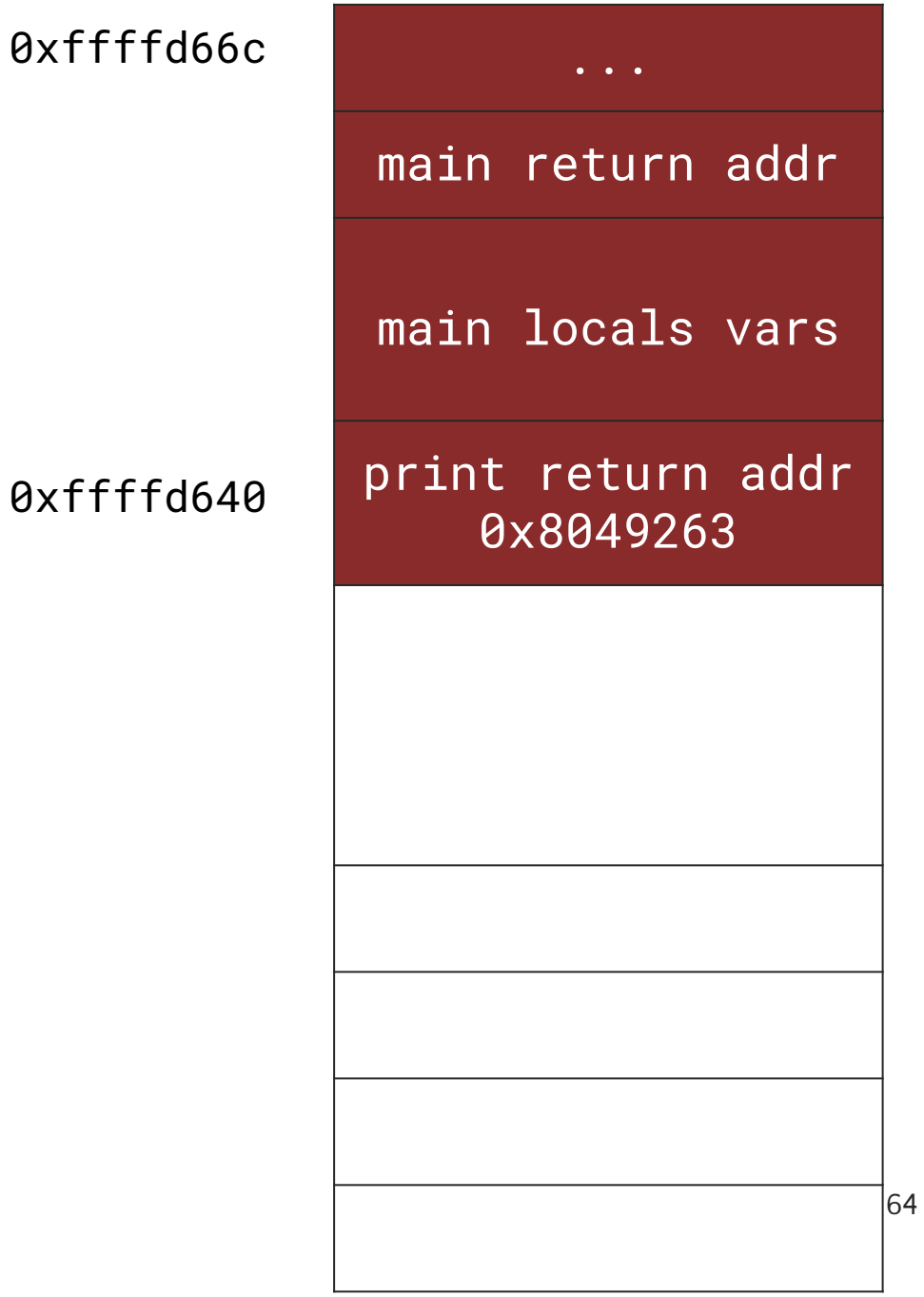
```
804925e:      e8 74 ff ff ff    call 80491d7 <print>
```

esp = 0xffffd640; eip = 0x804925e

A Toy Example

```
void print(char * message) {
    char buffer[64];
    strcpy(buffer, message);
    printf("%s\n", buffer);
}
int main(int argc, char ** argv) {
    ...
    print(argv[1]);
    return 0;
}
```

```
80491d7:      53      push   %ebx
esp = 0xffffd63c; eip = 0x80491d7
```



A Toy Example

```
void print(char * message) {
    char buffer[64];
    strcpy(buffer, message);
    printf("%s\n", buffer);
}
int main(int argc, char ** argv) {
    ...
    print(argv[1]);
    return 0;
}
```

```
8049202:    e8 69 fe ff ff   call   8049070 <puts@plt>
```

esp = 0xffffd5e0; eip = 0x8049202

A Toy Example

```
void print(char * message) {
    char buffer[64];
    strcpy(buffer, message);
    printf("%s\n", buffer);
}
int main(int argc, char ** argv) {
    ...
    print(argv[1]);
    return 0;
}
```

```
804920f:    c3                          ret
```

esp = 0xffffd63c; eip = 0x804920f

A Toy Example

```c
void print(char * message) {
    char buffer[64];
    strcpy(buffer, message);
    printf("%s\n", buffer);
}
int main(int argc, char ** argv) {
    ...
    print(argv[1]);
    return 0;
}
```

```
8049263:    83 c4 10                add $0x10,%esp
```

esp = 0xffffd640; eip = 0x8049263

Buffer Overflows

What Are Buffer Overflows?

A ***buffer overflow*** occurs when data is written <u>outside</u> of the space allocated for the buffer

- C does not check that writes are in-bounds

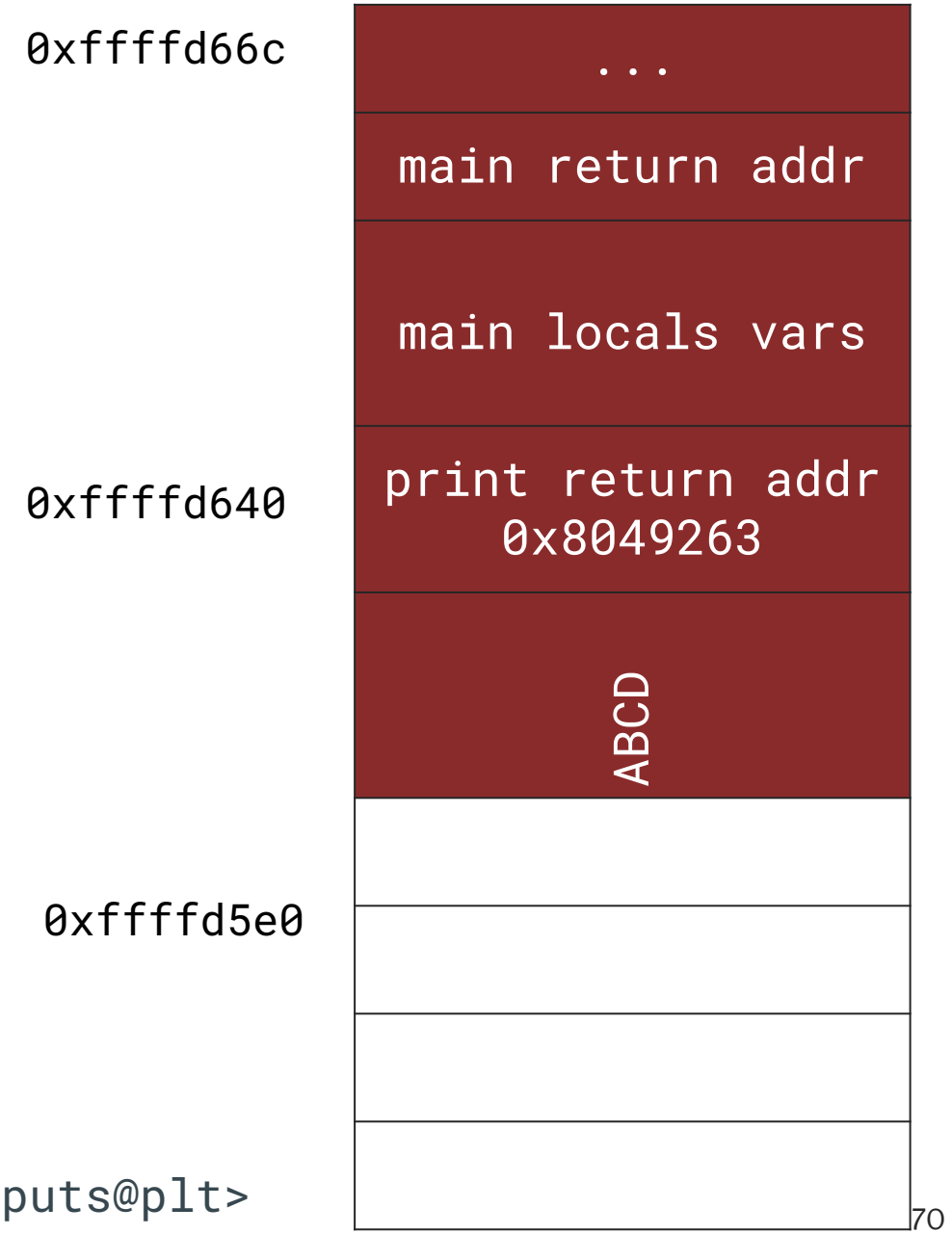
1. Stack-based
   - what we'll cover today
2. Heap-based
   - more advanced
   - very dependent on system and library version

A Toy Example

Input: ./bof0 "ABCD"

```
void print(char * message) {
    char buffer[64];
    strcpy(buffer, message);
    printf("%s\n", buffer);
}
int main(int argc, char ** argv) {
    ...
    print(argv[1]);
    return 0;
}
```

```
8049202:    e8 69 fe ff ff   call   8049070 <puts@plt>
esp = 0xffffd5e0; eip = 0x8049202
```

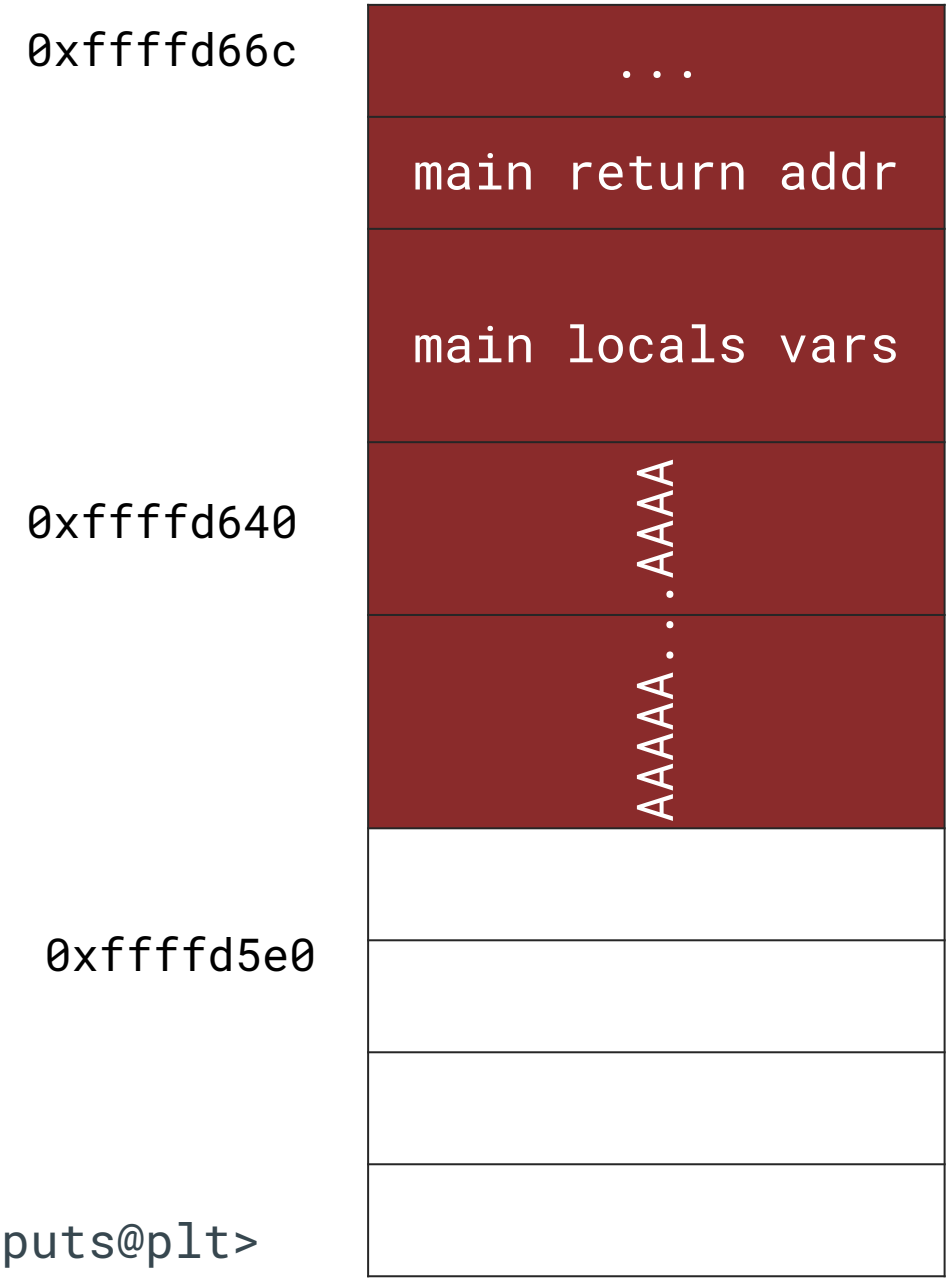


A Toy Example

Input: ./bof0 "AAAAAA…AAA"

```
void print(char * message) {
    char buffer[64];
    strcpy(buffer, message);
    printf("%s\n", buffer);
}
int main(int argc, char ** argv) {
    ...
    print(argv[1]);
    return 0;
}
```

```
8049202:    e8 69 fe ff ff   call   8049070 <puts@plt>
esp = 0xffffd5e0; eip = 0x8049202
```



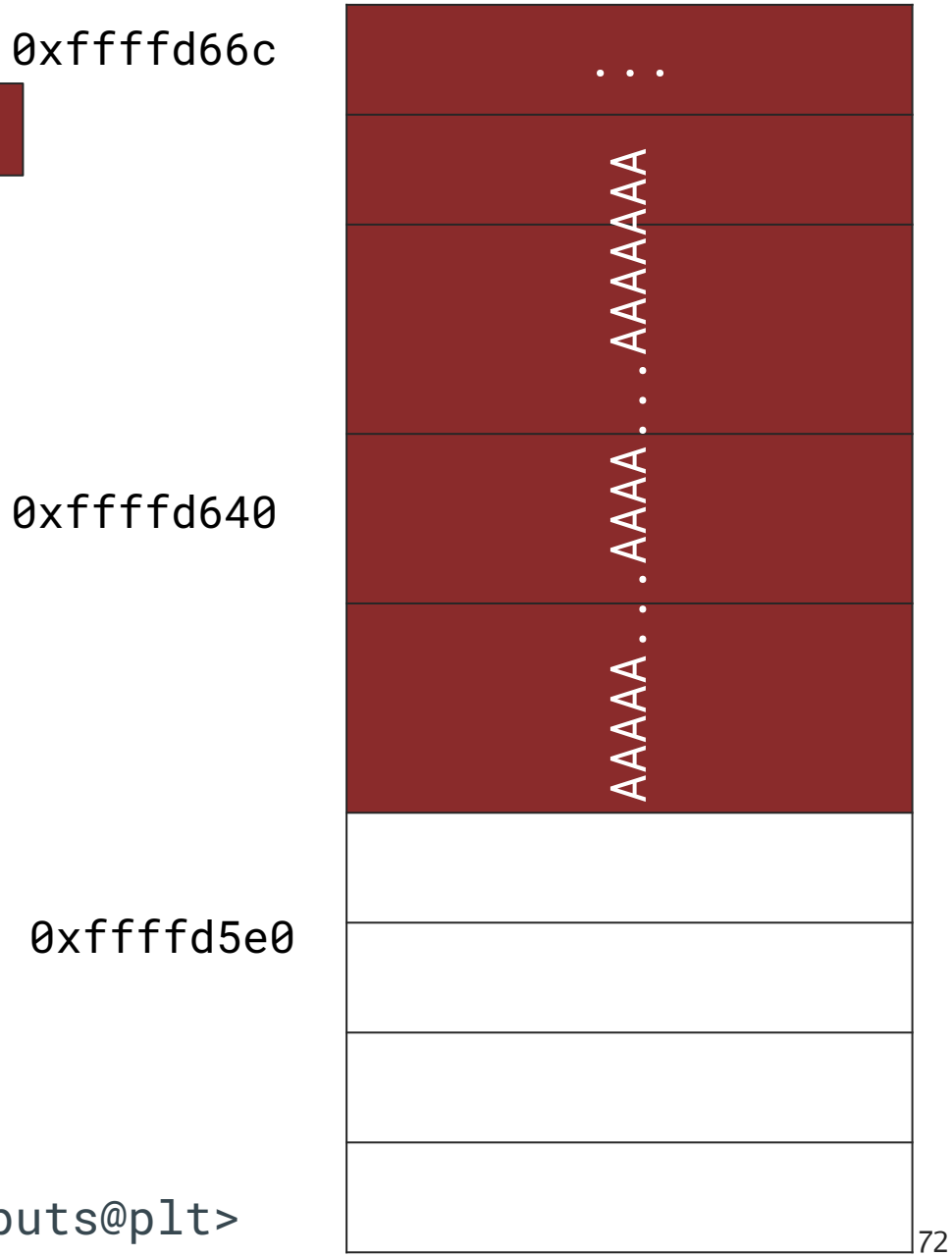
A Toy Example

Input: ./bof0 "AAAAAA…AAAAA…AAA"

```
void print(char * message) {
    char buffer[64];
    strcpy(buffer, message);
    printf("%s\n", buffer);
}
int main(int argc, char ** argv) {
    ...
    print(argv[1]);
    return 0;
}
```

```
8049202:    e8 69 fe ff ff   call   8049070 <puts@plt>
```

esp = 0xffffd5e0; eip = 0x8049202



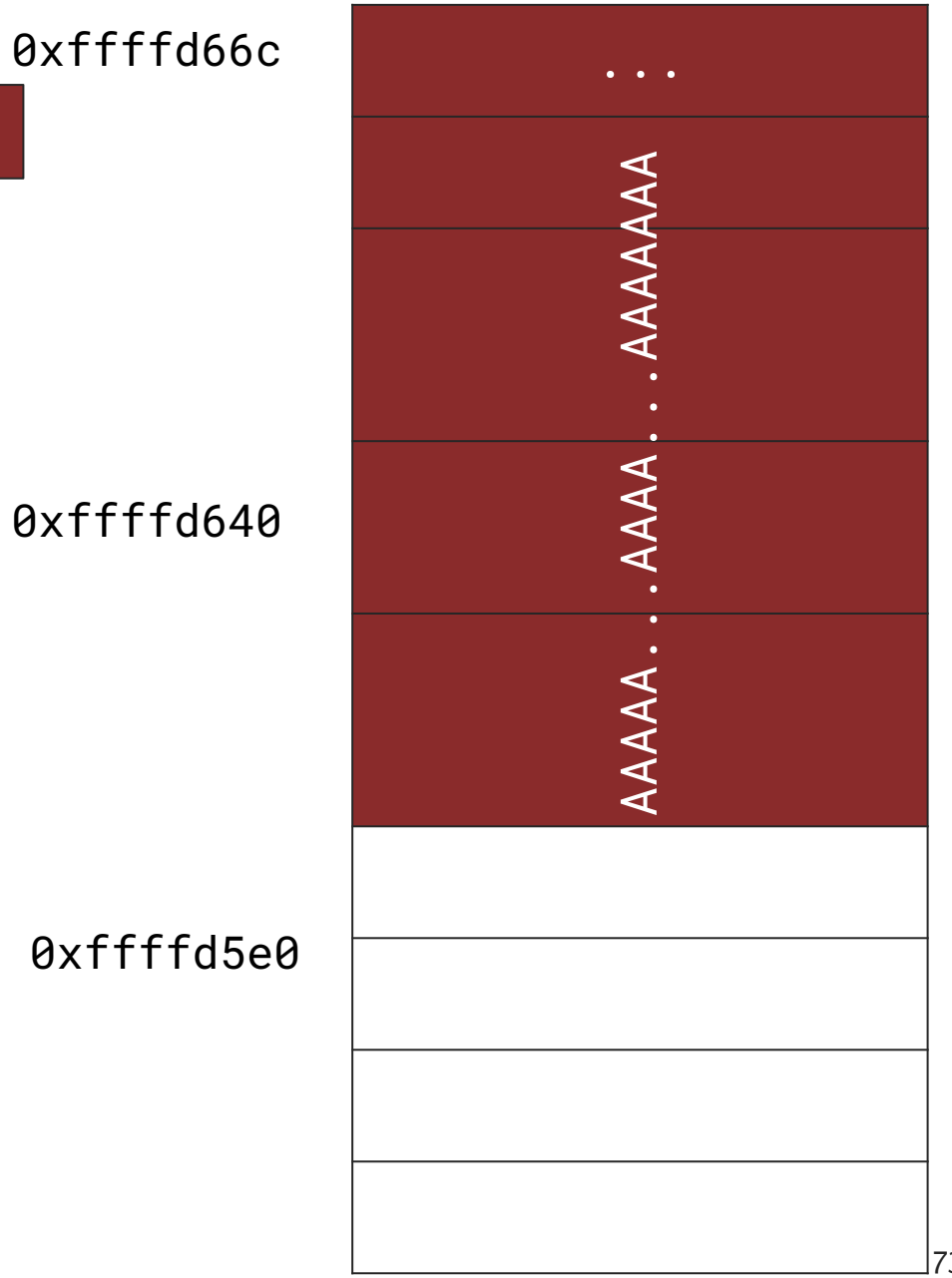
A Toy Example

Input: ./bof0 "AAAAAA…AAAAA…AAA"

```
void print(char * message) {
    char buffer[64];
    strcpy(buffer, message);
    printf("%s\n", buffer);
}
int main(int argc, char ** argv) {
    ...
    print(argv[1]);
    return 0;
}
```

```
804920f:    c3                          ret
```

esp = 0xffffd63c; eip = 0x804920f

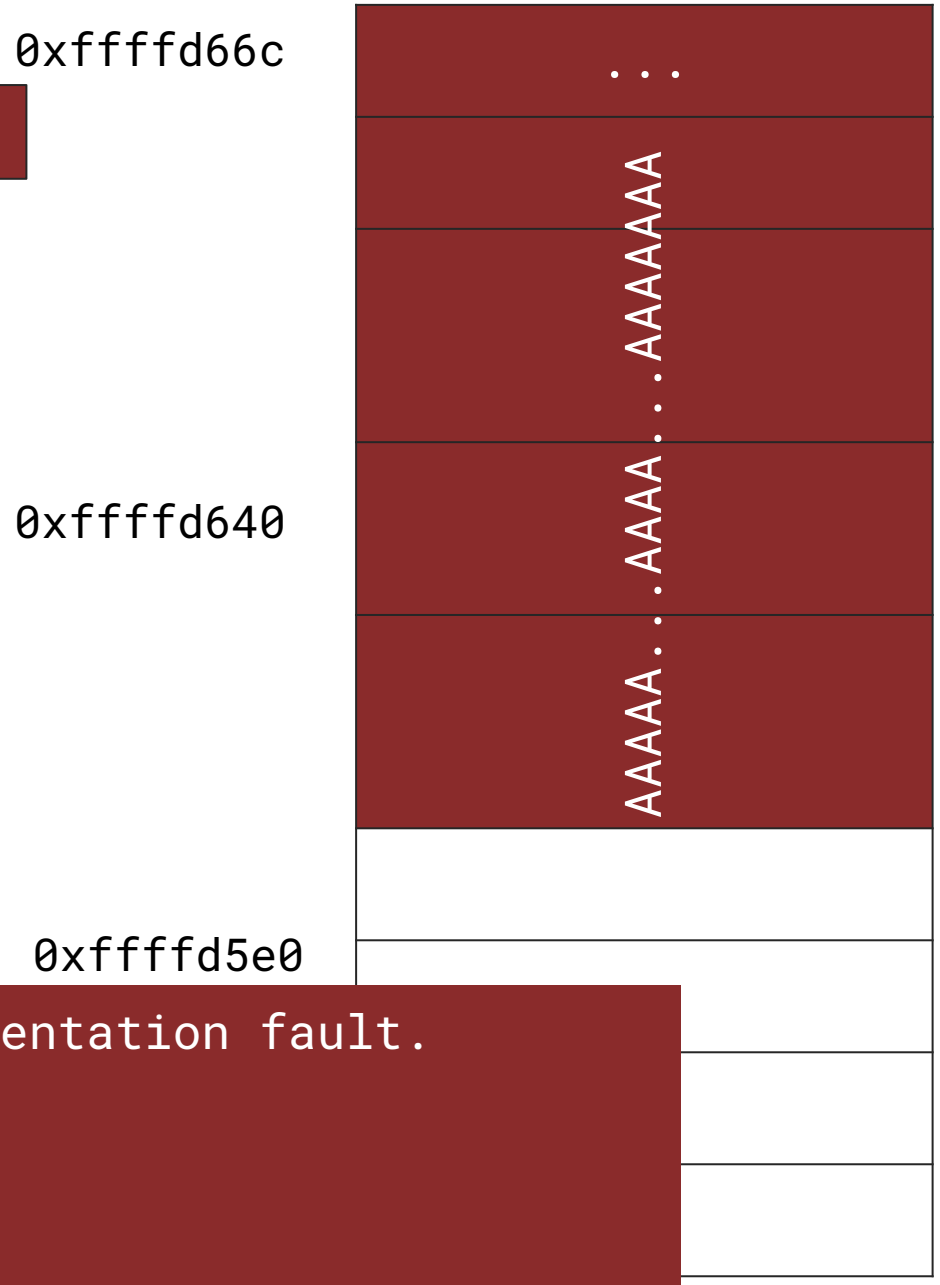


A Toy Example

Input: ./bof0 "AAAAAA…AAAAA…AAA"

```c
void print(char * message) {
    char buffer[64];
    strcpy(buffer, message);
    printf("%s\n", buffer);
}
int main(int argc, char ** argv) {
    ...
    print(argv[1]);
    return 0;
}
```

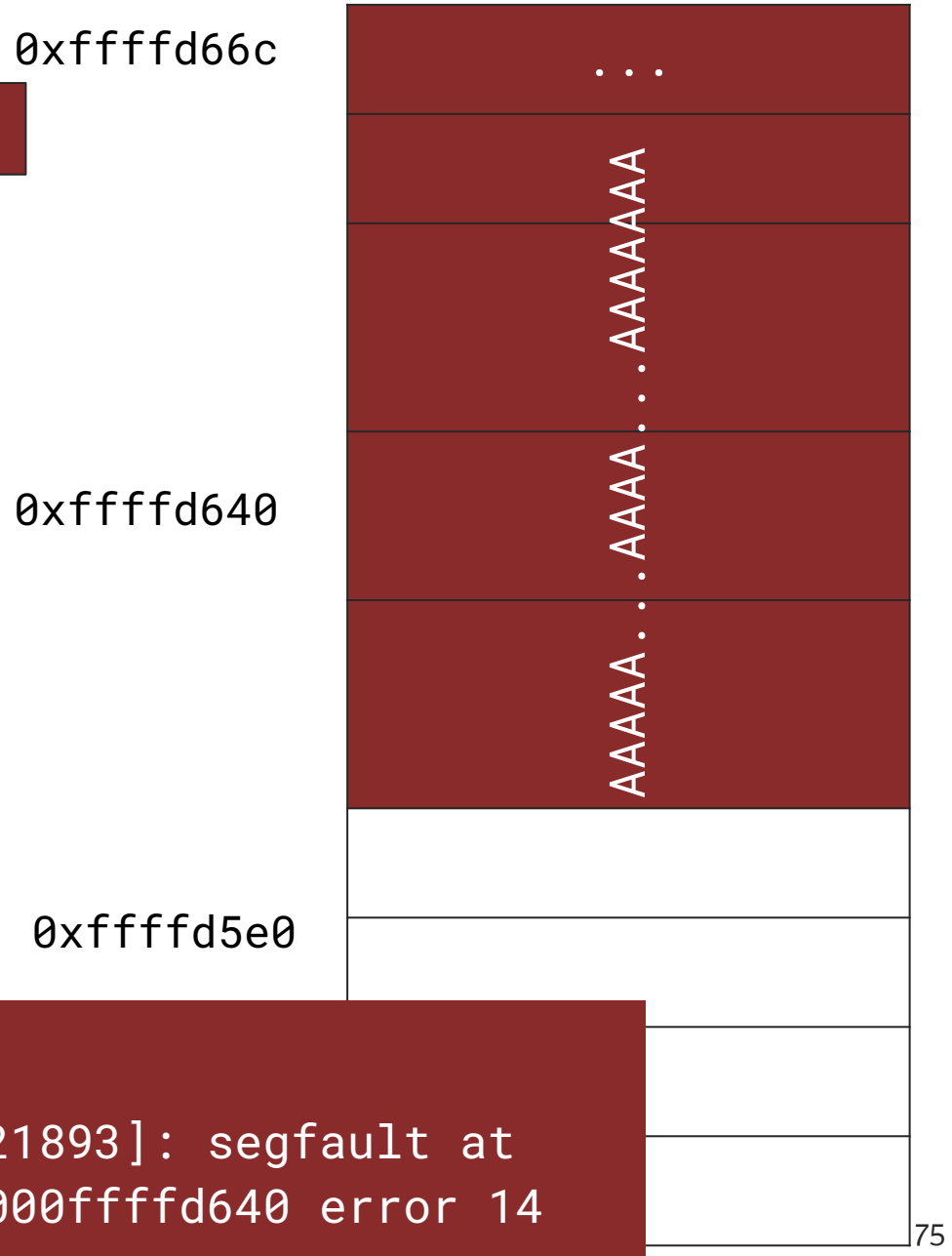
```
Program received signal SIGSEGV, Segmentation fault.
0x41414141 in ?? ()
(gdb) i r eip
eip            0x41414141       0x41414141
```



A Toy Example

Input: ./bof0 "AAAAAA…AAAAA…AAA"

```
void print(char * message) {
    char buffer[64];
    strcpy(buffer, message);
    printf("%s\n", buffer);
}
int main(int argc, char ** argv) {
    ...
    print(argv[1]);
    return 0;
}
```




```
$ dmesg -T | grep segfault
[Thu Mar 21 10:58:52 2024] bof0_real[21893]: segfault at
41414141 ip 0000000041414141 sp 00000000ffffd640 error 14
in libc.so.6[f7d85000+20000]
```

Can we return control to another function?

Let's do it Live!

I need 2-3 volunteers

```
ssh ubuntu@52.90.7.50
ilovedi@uoa
```

Shellcode Injection

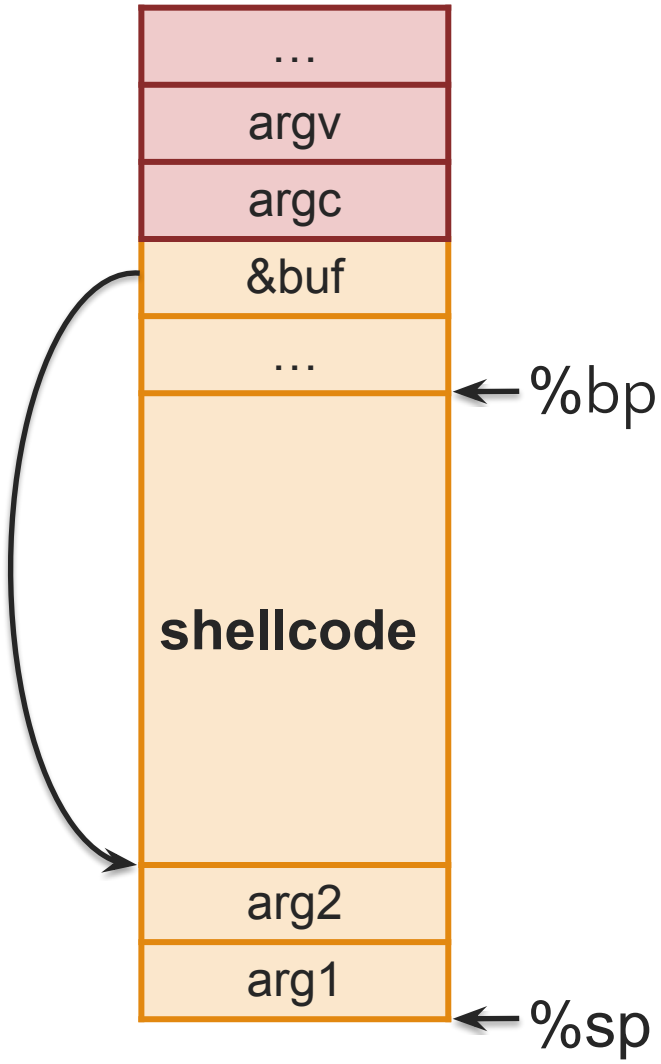
Shellcode

Traditionally exploits injected assembly instructions for `exec("/bin/sh")` into buffer.

Data Execution Prevention and other defenses have made this exploitation technique ineffective on consumer commercial OSes for over a decade.

Sadly, this is still applicable in areas like IoT, energy, and so on.

- Considered a basic skill for exploitation (even if not on your latest OS)
- See "*Smashing the stack for fun and profit*" for one string
- or search online OR *write it yourself*!

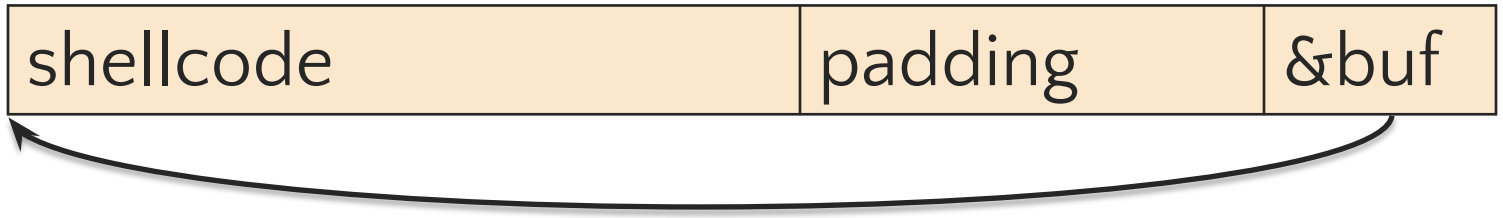


Recap

To generate ***exploit*** for a basic buffer overflow:

Determine size of stack frame up to head of buffer

1. Overflow buffer with the right size



computation + ***control***

How difficult is it to do that in reality?

Ευχαριστώ και καλή μέρα εύχομαι!

Keep hacking!